



Sublimation of ices of astrophysical interest: A bibliographic review

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ABSTRACT

The sublimation and condensation of ices play a very important role in the formation of planetary systems, in the evolution of some solar system bodies as well as in the equilibrium and matter exchanges between surface and atmosphere of most planets and satellites. The precise knowledge of vapour pressure of molecular solids at all relevant temperatures is mandatory, but the published sublimation relations are not always accurate enough. All published experimental measurements, and empirical and thermodynamical relations for the vapour pressure of 53 different species in their pure solid phases are reviewed. For several species, we also calculate the vapour pressure with accurate thermodynamic formulae from the triple point down to low temperatures. For 27 species (H_2O , O_2 , O_3 , CO , CO_2 , CH_3OH , HCOOH , CH_4 , C_2H_2 , C_2H_4 , C_2H_6 , C_6H_6 , HCN , HC_3N , C_2N_2 , C_4N_2 , N_2 , NH_3 , NO , N_2O , H_2S , SO_2 , AsH_3 , Ne , Ar , Kr , and Xe) we are able to propose vapour pressure relations, either empirical or theoretical, valid over a large range of temperatures, representative of astrophysical environments. All these relations are more accurate than those currently used in the astrophysical literature. In particular, most of the relations commonly used in the astrophysical literature are based on the data reported by Lide (2006) in the CRC Handbook of Chemistry and Physics, which are inaccurate for several compounds. The most problematic case is CO ice, for which a sublimation relation extrapolated from the liquid–gas equilibrium (Fanale and Salvail, 1990) is used in most of the models simulating the activity of comet nuclei.

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1. Introduction

Ices are ubiquitous in the solar system. They have been detected at the surface of most solar system bodies or in their atmospheres (Schmitt et al., 1998). Some of these bodies, like comet nuclei, KBOs, and icy satellites of the giant planets, contain large amounts of ices. These ices are not necessarily water ice and the most volatile species, such as N_2 , CO , and CH_4 , are not always trapped in amorphous water ice either. Indeed, pure or nearly pure ices of volatile species can exist at equilibrium on several objects of the outer solar system. For example, N_2 , CH_4 , H_2O , and CO have been detected at the surface of Pluto and Triton (Cruikshank et al., 1993; Owen et al., 1993; Quirico et al., 1999). On Pluto, some CH_4 and CO are incorporated in the almost pure ($> 99\%$) N_2 matrix, but there is also evidence for CH_4 in a pure state (Douté et al., 1999). Thus, the properties of pure ices, even for very volatile species such as CH_4 and N_2 , are of astrophysical interest.

At the very beginning of solar system formation most of the molecular species present in the solar nebula, except the most volatile species, such as H_2 and He , condensed during the cooling of the solar nebula (Yamamoto et al., 1983; Lewis, 1972). The subsequent sublimation of these species played, and still do play, a very important role in atmospheric formation. Cometary atmospheres are the most spectacular case since most of the volatiles that are observed today in cometary comae are produced by sublimation from the nuclei (Bockelée-Morvan et al., 2004; Espinasse et al., 1991; Squires and Beard, 1961). On some planets or satellites, ices and gas are involved in seasonal and climatic cycles. In these cases, energy is transported by sublimation and condensation through latent heat, with the bulk of gas flowing from regions receiving sunlight to areas receiving less or no solar energy, leading to cyclic seasonal transports of volatile ices. This is the case on Triton and Pluto, where dominant N_2 ice sublimates from the illuminated hemisphere, flows to and then condenses on the night hemisphere (Spencer et al., 1997; Grundy and Stansberry, 2000). On Mars a similar energy transport causes seasonal sublimation of CO_2 at one hemisphere and condensation at the surface of the opposite hemisphere. CO_2 being the main constituent of the Martian atmosphere, the total pressure of the atmosphere varies by about 30% over the seasonal cycle (Leighton and Murray, 1966; Encrenaz, 2001). On Io, the SO_2 atmosphere is

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probably due to sublimation of SO₂ ice patches or to plume sources (Trafton et al., 1998). Thus, the atmospheres of Pluto, Triton, Io, and Mars, despite having very different compositions and temperatures, are primarily supported by the vapour pressure of the dominant and most volatile ices on their surfaces. This is also certainly the case for most of the largest trans-Neptunian objects, where N₂-rich and CH₄ ices are detected (Merlin et al., 2009). In other atmospheres, the condensation of organic compounds and their sedimentation can act as a sink (Samuelson, 1998). This situation occurs in the atmosphere of Titan where numerous hydrocarbons and nitriles that are formed by photochemistry in the upper stratosphere and the ionosphere, condense in the form of aerosols or clouds in the lower stratosphere. The first Titan, condensate that has been clearly identified is C₄N₂ (Samuelson et al., 1997), but several other pure or mixed condensates, such as HCN, HC₃N, C₂H₂, C₂H₆, C₃H₈, and C₂H₅CN, could co-exist (Coustenis et al., 1999). In giant planets, numerous molecular species can also condense in crystalline form (Samuelson, 1998). In Uranus and Neptune atmospheres, some ice clouds of CH₄, H₂S, NH₄SH and H₂O could exist, although there is no direct spectroscopic identification for any of those condensates (de Pater et al., 1991). Even on Earth, H₂O condenses in the form of ice particles in cirrus and polar stratospheric clouds (Peter, 1998). On Mars, distinct CO₂ and H₂O ice clouds have also been identified.

Thus, the sublimation and condensation of ices play a very important role in the formation of planetary systems, in the evolution of some solar system bodies as well as in the equilibrium and matter exchanges between surface and atmosphere of most planets and satellites. Consequently, accurate knowledge of the sublimation–condensation curve of molecular solids at all relevant temperatures is mandatory to determine the various gas–solid equilibria and to model their exchanges on these objects. Unfortunately, as will be shown, the sublimation relations that are currently used in the astrophysical literature are not always accurate. The purpose of this article is to review the sublimation properties of pure ices of astrophysical interest. First, all the published experimental data concerning the sublimation pressure of compounds of astrophysical interest have been compiled. In a second step, relations to extrapolate to low temperatures are proposed, which could be easily used by the astrophysical community. This study deals with the sublimation–condensation equilibrium. For the liquefaction–vaporization equilibrium, we recommend the use of some other works such as Stull (1947a, 1947b), Boublik et al. (1984), and Lide (2006).

2. Thermodynamic and empirical relations of sublimation

The sublimation–condensation relations that we would like to propose should allow us to calculate, as accurately as possible, the sublimation pressure of studied species at any temperature. Indeed, for most species, experimental data have been measured at temperatures that are generally much too high to be representative of the astrophysical environments where these compounds are, or could be, present in the solid form. In this section, we present different ways to derive the sublimation curves at low temperatures. First, we describe the ‘thermodynamic calculation’, which is based on exact thermodynamic laws. Secondly, we present empirical relations used to interpolate and extrapolate experimental data and also results of the thermodynamic calculations. One should note that SI units have been used for all calculations presented in this article, but values of pressure are expressed in bar, which is the unit usually used in planetology.

2.1. Thermodynamic calculation

When a pure compound in gas phase is in equilibrium with its solid phase, the chemical potentials of both phases are equal. This equality leads to the Clausius–Clapeyron equation (see Eq. (1)), giving the slope of the equilibrium curve at a given temperature as a function of sublimation enthalpy ($\Delta H_{\text{sub}}(T)$ in J mol^{−1}) and molar volume of both phases (v in m³ mol^{−1}).

$$\left(\frac{dP_{\text{sub}}}{dT}\right)_{\text{eq}} = \frac{\Delta H_{\text{sub}}(T)}{T \cdot (v_{\text{gas}} - v_{\text{sol}})} \quad (1)$$

Although this equation is generally believed to be unsolvable without preliminary substitution of a simple expression for $\Delta H_{\text{sub}}(T)$ and $\Delta v(T) = v_{\text{gas}} - v_{\text{sol}}$, it has been demonstrated by Mosselman et al. (1982) that it can be transformed to an ‘exact differential equation’. The integration of the equation leads to thermodynamic relations, which can be used for calculations of vapour pressure and enthalpy of sublimation as a function of temperature. These relations are given in different forms in Gottschal and Korvezee (1953), Fisher and McMillan (1958), Brown and Ziegler (1979), Mosselman et al. (1982), and Feistel and Wagner (2007). Calculations of the vapour pressure curve make use of (1) a given point of the vapour pressure curve (generally the triple point), (2) heat capacity (C_p) and molar volume of the solid phase as a function of temperature, (3) sublimation enthalpy value at a given pressure and temperature (generally the triple point), (4) enthalpies and temperatures of all solid phases transitions, (5) heat capacity of the gas phase or thermodynamic functions of ideal gas calculated from statistical mechanics, and (6) an equation of state for the gas phase.

This thermodynamic approach allows the calculation of vapour pressure as a function of temperature without any particular hypothesis if all thermodynamic functions are known for the considered compound. This approach has been used by Brown and Ziegler (1979) to calculate the sublimation curves of H₂, O₂, N₂, F₂, CO, CO₂, Ne, Ar, Kr, Xe, CH₄, C₂H₄, and C₂H₆ and by Feistel and Wagner (2007) for H₂O.

Here, we suppose that the gas can be modelled as a ‘perfect gas’ and that the molar volume of the solid (v_{sol}) can be neglected against that of the gas phase (v_{gas}). Both hypotheses are good approximations since the calculation is performed at temperatures corresponding to low equilibrium pressures.

The Clausius–Clapeyron equation can then be written as

$$\left(\frac{dP_{\text{sub}}}{dT}\right)_{\text{eq}} = \frac{\Delta H_{\text{sub}}(T)P_{\text{sub}}}{RT^2} \quad (2)$$

For a perfect gas, we have $dH_{\text{gas}}(T) = C_{p,\text{gas}}(T) \cdot dT$, where $C_{p,\text{gas}}(T)$ is the molar heat capacity of the gas (in J K^{−1} mol^{−1}), and for ice we have $dH_{\text{sol}}(T) = C_{p,\text{sol}}(T) \cdot dT$. The sublimation enthalpy as a function of the temperature can then be expressed as

$$\Delta H_{\text{sub}}(T) = \int_{T_0}^T \Delta C_p(T) dT + \Delta H_{\text{sub}}(T_0) \quad \text{with} \quad \Delta C_p(T) = C_{p,\text{gas}}(T) - C_{p,\text{sol}}(T)$$

where T_0 is a reference temperature for which the sublimation enthalpy is known. Finally, the Clausius–Clapeyron equation can be integrated:

$$\ln(P_{\text{sub}}) = \int_{T_0}^T \frac{\Delta H_{\text{sub}}(T)}{RT^2} dT + \ln(P_{\text{sub}0}) \quad (3)$$

Knowing the sublimation pressure and enthalpy at a temperature T_0 , as well as the temperature dependence of $C_{p,\text{sol}}(T)$ and $C_{p,\text{gas}}(T)$, the sublimation pressure can be calculated at any temperature. Generally $C_{p,\text{sol}}(T)$ has been measured experimentally, but $C_{p,\text{gas}}(T)$ cannot be measured due to the low vapour pressure at low temperature. Nevertheless, $C_{p,\text{gas}}(T)$ can be

calculated thanks to the relation given by Vidler and Tennyson (2000), which used the internal partition function of the considered species. For some compounds, values of the internal partition function can be found in the HITRAN database (Rothman et al., 2005). This procedure was used by Feistel and Wagner (2007) to compute the sublimation pressure of water ice. The internal partition function in the HITRAN database is given only for temperatures higher than 70 K, so for lower temperatures, we extrapolated these data considering that this function is proportional to T for linear molecules and to $T^{1.5}$ for asymmetric molecules. The validity of this extrapolation has been checked thanks to the data at low temperature given by JPL (<http://spec.jpl.nasa.gov/>) and CDMS (<http://www.astro.uni-koeln.de/site/vorher sagen/>) databases.

In practise, we choose a reference temperature T_0 at which some experimental measurements are available. Then, knowing the heat capacities of the gas and the solid, we compute the sublimation pressure at any temperature.

To test this method, we have calculated the sublimation pressure of solid CO_2 considering only the experimental data of Giauque and Egan (1937), which range from 154 to 196 K. To perform this calculation, we use the heat capacity of ice measured by Giauque and Egan (1937), and the heat capacity of gas is calculated, thanks to the relation given by Vidler and Tennyson (2000), from the internal partition function of CO_2 molecules given in the HITRAN database. Temperature T_0 is chosen equal to 170 K. The resulting sublimation pressures are compared to the other available experimental data (Meyers and Van Dusen, 1933; Tickner and Lossing, 1951; Bryson et al., 1974; Fernandez-Fassnacht and Rio, 1984; Eiler et al., 2000) ranging from 70 to 216 K. We also compare this thermodynamic calculation with the usual empirical relation ($\ln(P_{\text{sub}})=A+B/T$) obtained considering only the experimental data of Giauque and Egan (1937). All results, including deviations of the calculations from the measured pressure, are presented in Fig. 1.

In the middle panel of Fig. 1, we can see that the thermodynamic calculation reproduces all experimental measurements with deviations between -50% and $+50\%$, whereas pressure measurements range over 12 orders of magnitude. These deviations may seem quite high, but they mostly reflect the dispersion of measurements at very low temperatures and pressures (below 10^{-7} bar). Moreover, the slope of this thermodynamic calculation seems to be more accurate as deviations from experimental data are equally distributed around zero. This is not the case for the empirical extrapolation, which overestimates all measurements at temperatures lower than 150 K. Thus, this thermodynamic method seems to be accurate. When the heat capacities are available, we extrapolate experimental measurements of sublimation pressure at low temperatures using this thermodynamic method.

2.2. Empirical extrapolations

Sometimes, the heat capacity of ice or gas is unknown, precluding the use of thermodynamic calculations. Then, it is necessary to use an empirical relation to interpolate the existing experimental data and to extrapolate them to low temperatures. In the literature, numerous empirical relations derived from the Clausius–Clapeyron equation can be found assuming that:

- (i) the gas is perfect,
- (ii) the molar volume of the condensed phase is negligible against that of the gas, and they differ by assuming different variations of the sublimation enthalpy with temperature
- If the sublimation enthalpy is supposed to be independent of temperature; then the sublimation pressure follows the

relation $\ln(P_{\text{sub}})=A+B/T$, which is the most common and simplest empirical relation.

- If the sublimation enthalpy is supposed to vary linearly with temperature, the sublimation pressure follows the relation $\ln(P_{\text{sub}})=A+B/T+C\ln(T)$ which is generally called the ‘Dupré’ or the ‘Rankine’ relation.
- If the sublimation enthalpy is supposed to vary as a second-order function of temperature, the sublimation pressure follows the relation $\ln(P_{\text{sub}})=A+B/T+C\ln(T)+DT$.

A few other empirical expressions are used. In particular the sublimation pressure is sometimes extrapolated using polynomials such as $\ln(P_{\text{sub}})=A_0+\sum_{i=1}^n A_i/T^i$.

Good interpolations of experimental data are obtained by using these empirical relations but for extrapolation some of them are inadequate, and can even present a slope change at low temperature, which has no physical meaning. All the empirical relations have been applied to the experimental CO_2 measurements of Giauque and Egan (1937) and compared to the measurements of the CO_2 sublimation pressure (Fig. 2). As can be seen in Fig. 2, the extrapolation, which used a polynomial function of the third order, presents a change of slope at about 50 K, which has no physical meaning. All empirical relations, except the first one ($\ln(P_{\text{sub}})=A+B/T$), can present this type of behaviour. As coefficient B is always negative, the last empirical relation ($\ln(P_{\text{sub}})=A+B/T$) is the only permanently increasing function with temperature. Thus, when the thermodynamic method cannot be used, we choose to extrapolate the experimental data with this simple empirical relation. In some cases, we also propose a more accurate interpolation of experimental data with higher-order polynomials, but this is only valid in the temperature range of these data.

These high-order polynomials are also used to accurately interpolate the results of the thermodynamic calculations between the triple point and the lowest temperature calculated (typically at 70 K). However, for the same reason as above, the polynomial relations should not be used as an extrapolation relation at lower temperatures.

3. Bibliographic review

3.1. Presentation of the review

We have carried out a survey of experimental measurements of the sublimation pressure of 53 species that are present or could be present in astrophysical environments. We have found data in the literature for 30 species and compiled them. The purpose of this review is to provide analytical relations, allowing easy and accurate calculation of the sublimation pressure at very low temperature. Table 1 lists the 53 species for which we have looked for sublimation pressure measurements together with previous review articles and books (Stull, 1947a, 1947b; Honig and Hook, 1960; Brown and Ziegler, 1979; Lide, 2006) in which sublimation pressure data have already been compiled, as well as articles of planetological interest in which sublimation relations have been provided (Yamamoto et al., 1983; Prialnik et al., 2004; Huebner et al., 2006). Table 1 gives the sub-section number in which some comments on individual species are provided and can be used as an index. Table 2 summarizes, for most of the studied species, the values of the triple point temperature and pressure found in the NIST Chemistry WebBook (National Institute of Standards and Technology, <http://webbook.nist.gov/chemistry/>), the database of AirLiquide (<http://encyclopedia.airliquide.com/encyclopedia.asp>), and the 87th edition of the CRC Handbook of Chemistry and Physics (Lide, 2006). Table 3 gives general information on the

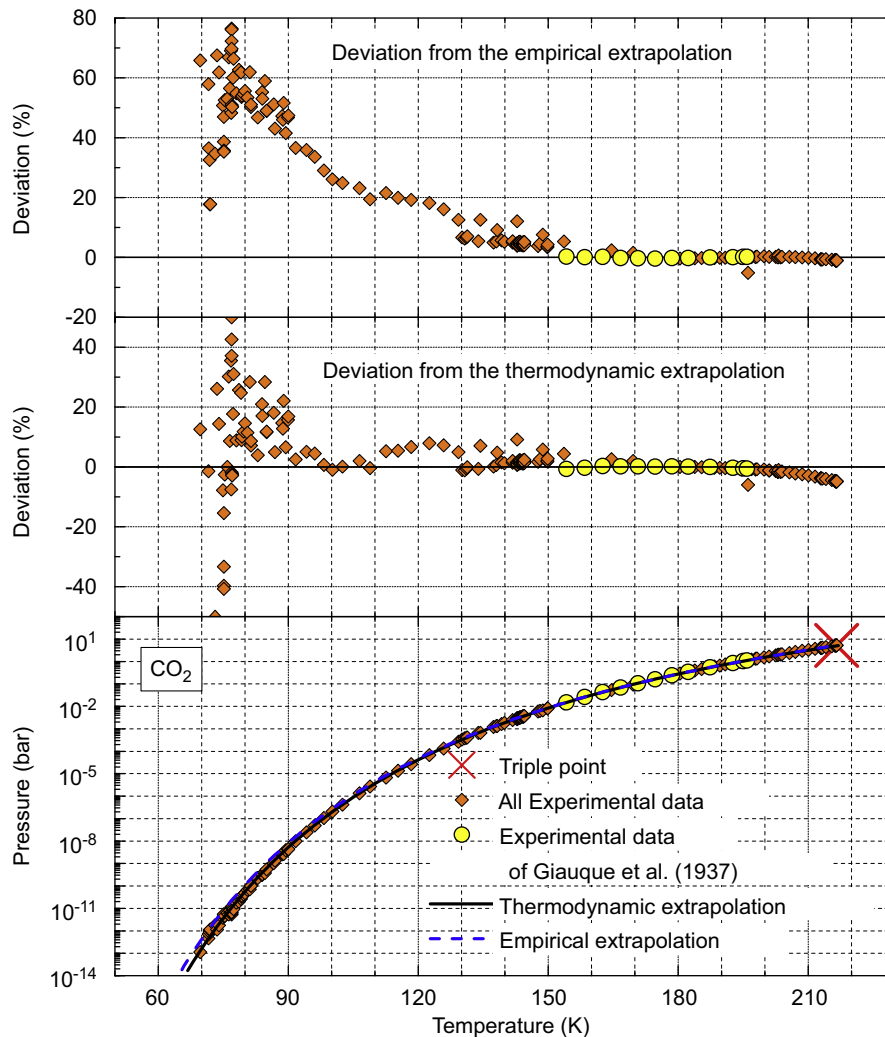


Fig. 1. Lower panel represents the sublimation pressure of CO₂ as a function of temperature. All the experimental data that we have compiled (Meyers and Van Dusen, 1933; Tickner and Lossing, 1951; Bryson et al., 1974; Fernandez-Fassnacht and Rio, 1984; Eiler et al., 2000) are represented as diamonds, except those measured by Giauque and Egan (1937), which are represented as circles. The thermodynamic and empirical extrapolations are represented as solid and dashed lines, respectively. Middle and upper panels represent the deviation of the thermodynamic and empirical extrapolations, respectively, from the experimental data.

experiments (temperature and pressure ranges of measurements, number of experimental points) in all used bibliographic sources.

For each solid crystalline phase of each species, we fit a polynomial expression of the form

$$\ln(P_{\text{sub}}) = A_0 + \sum_{i=1}^n A_i/T^i \quad (4)$$

to the sublimation pressure data over the temperature range of stability of the phase. Full polynomial relations (noted 'T', for interpolated thermodynamic relations, in Table 4) have been determined for data computed by the thermodynamic approach described in Section 2.1. When this thermodynamic approach is not possible (see Section 2.2), the simple empirical relation $\ln(P_{\text{sub}}) = A_0 + A_1/T$ (denoted by 'E', for empirical extrapolation, in Table 4) is used to fit experimental measurements. This provides the most reliable extrapolation of data at lower temperature. In two cases (NO and N₂O) we also used an empirical interpolation (denoted by 'I', for interpolation, in Table 4) of experimental measurements with a high-order polynomial. Detailed informations on these fits, especially deviations from the experimental data, are given in Table 4. Table 5 provides the coefficients of the polynomial relations. These coefficients are

generally given with 4 digits (15 for H₂O) because some negative and positive terms in the polynomials are almost equal and thus need high precision coefficients to preserve the accuracy of the thermodynamic calculations.

In the next section, we discuss available data for each species and accuracy of the proposed thermodynamic or empirical relations. Note that these new relations are always more accurate than the empirical ones currently used in astrophysical literature.

3.2. Comments on individual species

3.2.1. H₂O (water)

The triple point of H₂O is located at 273.16 K and $(6.11657 \pm 0.0001) \times 10^{-3}$ bar (Wagner and Pruss (2002) and Table 2). We can note that the temperature at the triple point of H₂O corresponds by definition to one of the defined fixed point of the temperature scale ITS-90. We have collected 216 experimental measurements ranging from 130 K (4.78×10^{-13} bar) to 273.15 K (6.11×10^{-3} bar; Douslin and Osborn, 1965; Jancso et al., 1970; Bryson et al., 1974; Marti and Mauersberger, 1993; White et al., 1998; Mauersberger and Krankowsky, 2003; Shilling et al., 2006; Table 3 and Fig. 3). Water ice can have crystalline or amorphous

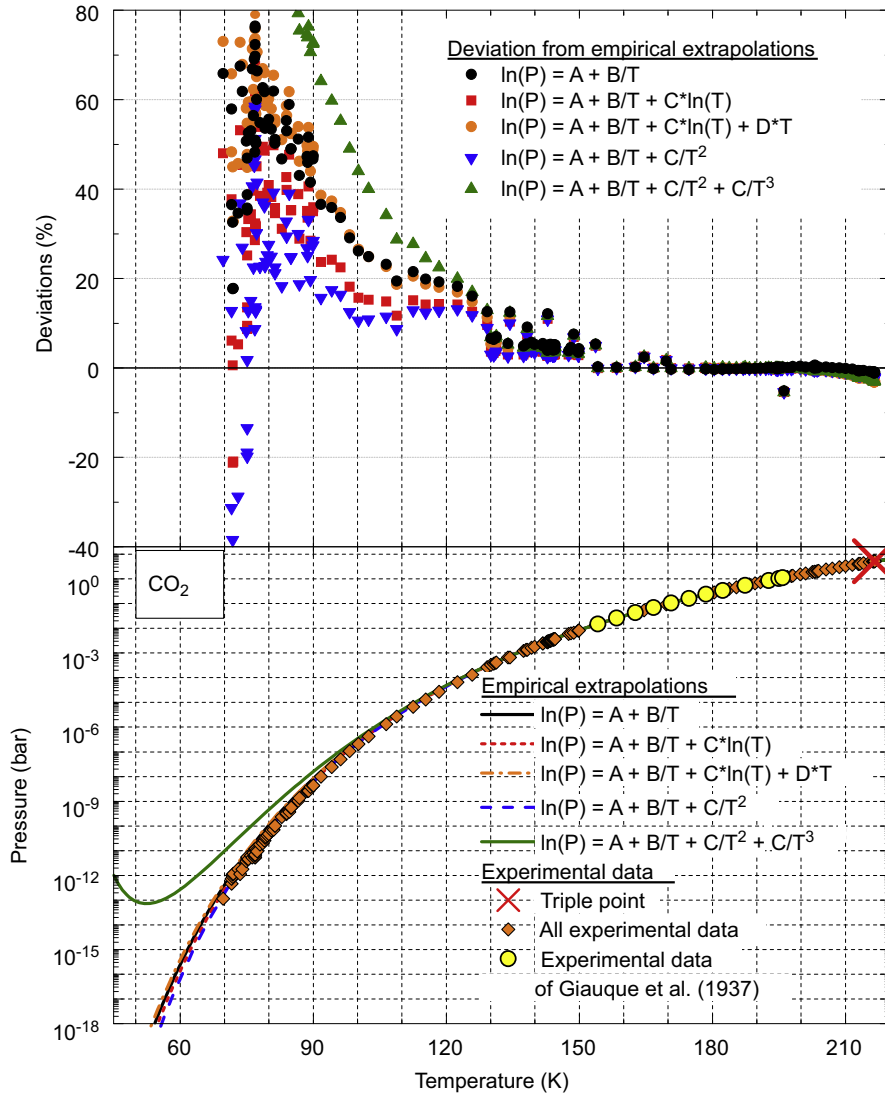


Fig. 2. Lower panel represents the sublimation pressure of CO₂ as a function of temperature. All the experimental data that we have compiled (Meyers and Van Dusen, 1933; Tickner and Lossing, 1951; Bryson et al., 1974; Fernandez-Fassnacht and Rio, 1984; Eiler et al., 2000) are represented as diamonds, except those measured by Giauque and Egan (1937), which are represented as circles. All the empirical relations that we have used to extrapolate the data of Giauque and Egan (1937) are represented by lines of different colours. Upper panel represents the deviation of the different empirical extrapolations from the experimental data. For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.

structure. In the studies of Marti and Mauersberger (1993) and Mauersberger and Krankowsky (2003), water ice has been deposited very slowly between 200 and 225 K. Under these conditions, water ice is almost certainly crystalline hexagonal ice (Ih). As Douslin and Osborn (1965) and Jancso et al. (1970) have worked at relatively high temperatures, their ice samples were also probably crystalline. Shilling et al. (2006) have checked the state of their ice samples by X-ray diffraction and performed 3 measurements on crystalline hexagonal ice (Ih). In the other studies (Bryson et al., 1974; White et al., 1998), it is not clear if the ice was amorphous or crystalline. Lide (2006) reports data for the sublimation equilibrium, which are in agreement with the previous experimental measurements (Fig. 3). Numerous relations have been published in the literature to compute the vapour pressure of hexagonal water ice (Washburn, 1924; Jancso et al., 1970; Yamamoto et al., 1983; Wagner et al., 1994; Murphy and Koop, 2005; Prialnik et al., 2004; Huebner et al., 2006; Feistel and Wagner, 2006, 2007). We have compared all these relations to the experimental data. The most complete and precise work has been published by Feistel and Wagner (2007). They have used the most recent data (measurements of vapour pressure and heat

capacity of hexagonal ice) to compute the vapour pressure of hexagonal water ice using an accurate thermodynamic approach. If we consider only the measurements of Marti and Mauersberger (1993) and Mauersberger and Krankowsky (2003), which have been obtained on crystalline ice, the deviation, between experimental measurements and this theoretical relation, reaches -27% and $+15\%$ (Fig. 3). This mostly reflects experimental dispersion of the measurements of Mauersberger and Krankowsky (2003) although if we consider all compiled experimental data, deviations vary between -40% and $+180\%$ (Fig. 3). The high vapour pressure values are possibly due to the presence of amorphous ice in the samples condensed below 140 K by Bryson et al. (1974) and White et al. (1998). Indeed it has already been shown by numerous authors (Kouchi, 1987; Sack and Baragiola, 1993) that vapour pressure of amorphous ice is higher by one or two orders of magnitude than for crystalline ice. All these authors have shown that vapour pressure for amorphous ice strongly depends on condensation temperature and on the rate of condensation. Therefore, no sublimation relation for amorphous water ice can be proposed as it does not depend only on temperature.

Table 1

Data on sublimation or evaporation pressures and extrapolation relations available in review articles and books.

Sub-section ^a	Species	Stull (1947a, 1947b)	Honig and Hook (1960)	Brown and Ziegler (1979)	Lide (2006)	Yamamoto et al. (1983)	Prialnik et al. (2004)	Huebner et al. (2006)
1	H ₂ O	X	X	–	X	X	X	X
2	H ₂ O ₂	X	–	–	X	–	–	–
3	O ₂	X	X	X	X	–	–	–
4	O ₃	X	–	–	X	–	–	–
5	CO	X	X	X	X	X	X	X
6	CO ₂	X	X	X	X	X	X	X
7	H ₂ CO	X	–	–	X	–	–	–
8	CH ₃ OH	X	–	–	X	–	X	–
9	HCOOH	X	–	–	X	–	–	–
10	HCOOCH ₃	X	–	–	X	–	–	–
11	HOCH ₂ CH ₂ OH	X	–	–	X	–	–	–
12	CH ₄	X	X	X	X	X	X	X
13	C ₂ H ₂	X	–	–	X	X	X	X
14	C ₂ H ₄	X	–	X	X	–	–	–
15	C ₂ H ₆	X	–	X	X	–	X	–
16	C ₃ H ₄ (a)	X	–	–	X	–	X	–
17	C ₃ H ₄ (p)	X	–	–	X	–	X	–
18	C ₃ H ₆	X	–	–	X	–	–	–
19	C ₃ H ₈	X	–	–	X	–	–	–
20	C ₄ H ₂	X	–	–	–	–	–	–
21	C ₆ H ₂	–	–	–	–	–	–	–
22	C ₆ H ₆	X	–	–	X	–	–	–
23	HCN	X	–	–	X	X	X	X
24	HNC	–	–	–	–	–	–	–
25	HC ₃ N	–	–	–	X	–	–	–
26	CH ₃ CN	X	–	–	X	–	–	–
27	CH ₃ CH ₂ CN	X	–	–	X	–	–	–
28	C ₂ N ₂	X	–	–	X	–	–	X
29	C ₄ N ₂	–	–	–	–	–	–	–
30	N ₂	X	X	X	X	X	–	X
31	NH ₃	X	X	–	X	X	X	X
32	NO	X	X	–	X	–	–	X
33	NO ₂	–	–	–	–	–	–	–
34	N ₂ O	X	X	–	X	–	–	X
35	HNCO	–	–	–	–	–	–	–
36	NH ₂ CHO	X	–	–	X	–	–	–
37	H ₂ S	X	X	–	X	–	X	X
38	SO ₂	X	X	–	X	–	–	–
39	CS ₂	X	X	–	X	–	–	–
40	OCS	X	X	–	X	–	–	–
41	H ₂ SO ₄	X	–	–	X	–	–	–
42	NH ₄ SH	X	–	–	–	–	–	–
43	H ₂ CS	–	–	–	–	–	–	–
44	Cl ₂	X	X	–	X	–	–	–
45	ClS ₂	–	–	–	–	–	–	–
46	ClO	–	–	–	–	–	–	–
47	PH ₃	X	–	–	X	–	–	–
48	AsH ₃	X	–	–	–	–	–	–
49	SiH ₄	X	–	–	X	–	–	–
50	Ne	X	X	X	X	–	–	X
51	Ar	X	X	X	X	–	–	X
52	Kr	X	X	X	X	–	–	X
53	Xe	X	X	X	X	–	–	X

^a Numbers quoted in the first column correspond to the subsection of Section 3.2 in which we review in detail the data for each species.

We can also note that under certain conditions, water can form crystalline cubic ice (Ic). Only 3 experimental measurements of vapour pressure of the cubic phase have been performed over the temperature range 180–190 K (Shilling et al., 2006; Fig. 3). The authors have determined that the value is $10.5\% \pm 0.3\%$ higher than the vapour pressure of the hexagonal phase. We can assume that the vapour pressure of Ic ice is 10.5% higher than for Ih ice at all temperatures where this phase is stable but this hypothesis has to be confirmed by experimental measurements at $T < 180$ K and $T > 190$ K. No other vapour pressure relation for Ic ice can be proposed for now.

In astrophysical models, different empirical sublimation relations are used. They have been proposed by Yamamoto et al. (1983), Fanale and Salvail (1984) (which is used by Prialnik et al.,

2004), and Huebner et al. (2006). These relations present higher deviations from the experimental data than those computed by Feistel and Wagner (2007). We recommend the use of the thermodynamic relation computed by Feistel and Wagner (2007), which can be expressed by an empirical function (see Eqs. (5) and (6) and parameter values in Table 6). In both equations, T_t and P_t stand for the temperature and pressure at the triple point, respectively. Both values are given in Table 2.

$$\ln\left(\frac{P_{\text{subl}}(T)}{P_t}\right) = \frac{3}{2} \ln\left(\frac{T}{T_t}\right) + \left(1 - \frac{T_t}{T}\right) \eta\left(\frac{T}{T_t}\right) \quad (5)$$

$$\eta\left(\frac{T}{T_t}\right) = \sum_{i=0}^6 e_i \left(\frac{T}{T_t}\right)^i \quad (6)$$

Table 2
Triple point determinations.

Species	NIST Webbook		Air Liquide		CRC Handbook (Lide, 2006)		Others references		Selected	
	T (K)	P (bar)	T (K)	P (bar)	T (K)	P (bar)	T (K)	P (bar)	T (K)	P (bar)
H ₂ O	273.16	$(0.0061 \pm 0.0001) \times 10^{-3}$	–	–	273.16	6.1166×10^{-3}	273.16 ^a	$(6.11657 \pm 0.0001) \times 10^{-3a}$	273.16	$(6.11657 \pm 0.0001) \times 10^{-3}$
O ₂	54.33 ± 0.06	–	54.35	1.52×10^{-3}	–	–	54.352 ^b	1.46×10^{-3b}	54.33 ± 0.06	$(1.49 \pm 0.03) \times 10^{-3}$
O ₃	–	–	80.6	1.14×10^{-5}	–	–	79.6 ± 0.3 ^c	$(5.6 \pm 0.6) \times 10^{-6c}$	79.6 ± 0.3	$(5.6 \pm 0.6) \times 10^{-6}$
CO	68.03 ± 0.14	–	68.14	0.1535	68.13	0.154	–	–	68.1 ± 0.1	0.1537 ± 0.0003
CO ₂	216.58 ± 0.03	5.185 ± 0.005	216.58	5.185	216.58	5.18	–	–	216.58 ± 0.03	5.185 ± 0.005
H ₂ CO	155.1 ± 0.3	–	–	–	–	–	–	–	155.1 ± 0.3	$(4 \pm 1) \times 10^{-4d}$
CH ₃ OH	175.5 ± 0.5	–	–	–	–	–	–	–	175.5 ± 0.5	$(1.8 \pm 0.2) \times 10^{-6d}$
HCOOH	281.42 ± 0.07	0.0236 ± 0.0001	–	–	–	–	–	–	281.42 ± 0.07	0.0236 ± 0.0001
HOCH ₂ CH ₂ OH	258.8 ± 2.4	–	–	–	–	–	–	–	258.8 ± 2.4	–
CH ₄	90.67 ± 0.03	0.1169 ± 0.0006	90.68	0.117	90.69	0.117	90.6854 ± 0.0003 ^e	0.11696 ± 0.00002 ^e	90.6854 ± 0.0003	0.11696 ± 0.00002
C ₂ H ₂	192 ± 1	1.2825 ± 0.004	192.60	1.282	192.4	1.26	–	–	192 ± 1	1.275 ± 0.015
C ₂ H ₄	104.0 ± 0.1	0.0012 ± 0.0001	103.97	0.0012	–	–	–	–	104.0 ± 0.1	0.0012 ± 0.0001
C ₂ H ₆	91 ± 6	$(1.1 \pm 0.1) \times 10^{-5}$	89.24	1.1×10^{-5}	–	–	90.34 ± 0.01 ^f	$(1.1 \pm 0.1) \times 10^{-5f}$	90.34 ± 0.01	$(1.1 \pm 0.1) \times 10^{-5}$
C ₃ H ₄ (a)	136.59 ± 0.20	–	–	–	–	–	–	–	136.59 ± 0.20	$(9.3 \pm 0.4) \times 10^{-5d}$
C ₃ H ₄ (p)	168.5 ± 0.3	–	–	–	–	–	–	–	168.5 ± 0.3	$(3.9 \pm 0.2) \times 10^{-3d}$
C ₃ H ₆	87.8 ± 0.8	–	–	–	–	–	–	–	87.8 ± 0.8	$10^{-7-10^{-8d}}$
C ₃ H ₈	85 ± 3	$(1.687 \pm 0.003) \times 10^{-9}$	–	–	–	–	85.47 ± 0.05 ^f	$(1.685 \pm 0.001) \times 10^{-9f}$	85.47 ± 0.05	$(1.685 \pm 0.001) \times 10^{-9}$
C ₄ H ₂	–	–	–	–	–	–	–	–	235.6 ± 3.5 ^d	0.1 ± 0.03 ^d
C ₆ H ₆	278.5 ± 0.6	–	–	–	–	–	278.68 ± 0.02 ^g	$(4.785 \pm 0.005) \times 10^{-2g}$	278.68 ± 0.02	$(4.785 \pm 0.005) \times 10^{-2}$
HCN	259.86 ± 0.02	–	259.85	0.187	259.83	0.1862	–	–	259.85 ± 0.03	0.1866 ± 0.0004
CH ₃ CN	229.32 ± 0.02	–	–	–	–	–	–	–	229.32 ± 0.02	$(2.0 \pm 0.1) \times 10^{-3d}$
CH ₃ CH ₂ CN	180.37 ± 0.02	–	–	–	–	–	–	–	180.37 ± 0.02	$(3.8 \pm 0.1) \times 10^{-6d}$
C ₂ N ₂	245.32 ± 0.02	0.751 ± 0.013	245.32	0.738	–	–	–	–	245.32 ± 0.02	0.744 ± 0.009
N ₂	63.14 ± 0.06	0.1255 ± 0.0005	63.148	0.1253	–	–	–	–	63.14 ± 0.06	0.1255 ± 0.0005
NH ₃	–	0.06060 ± 0.00005	195.41	0.06077	195.4	0.0612	–	–	195.41 ± 0.01	0.0609 ± 0.0003
NO	–	–	–	–	109.5	0.219	–	–	109.5 ± 0.1	0.219 ± 0.001
N ₂ O	–	0.8790 ± 0.0001	182.34	0.878	–	–	182.30 ± 0.05 ^h	0.879 ± 0.001 ^h	182.30 ± 0.05	0.879 ± 0.001
NH ₂ CHO	275.66 ± 0.8	–	–	–	–	–	–	–	275.66 ± 0.8	–
H ₂ S	187.65 ± 0.07	0.232 ± 0.005	187.45	0.227	187.6	0.227	–	–	187.57 ± 0.15	0.229 ± 0.008
SO ₂	197.64 ± 0.05	0.0167 ± 0.0001	197.63	0.0167	–	–	–	–	197.64 ± 0.05	0.0167 ± 0.0001
CS ₂	161.45 ± 0.40	–	–	–	–	–	–	–	161.45 ± 0.40	$(2 \pm 1) \times 10^{-5d}$
OCS	134.32 ± 0.20	–	–	–	–	–	–	–	134.32 ± 0.20	$(5 \pm 2) \times 10^{-4d}$
Cl ₂	172.12 ± 0.15	$(1.390 \pm 0.003) \times 10^{-2}$	172.15	1.4×10^{-2}	–	–	–	–	172.12 ± 0.15	$(1.39 \pm 0.01) \times 10^{-2}$
PH ₃	139.41 ± 0.20	–	139.35	3.6×10^{-2}	–	–	–	–	139.41 ± 0.20	$(3.6 \pm 0.1) \times 10^{-2}$
AsH ₃	–	–	156.25	0.03	–	–	156.23 ± 0.02 ⁱ	$(2.98 \pm 0.02) \times 10^{-2i}$	156.23 ± 0.02	$(2.98 \pm 0.02) \times 10^{-2}$
Ne	24.56 ± 0.05	0.433 ± 0.003	24.55	0.433	–	–	–	–	24.56 ± 0.05	0.433 ± 0.003
Ar	83.79 ± 0.06	0.689 ± 0.001	83.78	0.687	83.81	0.688	83.8058 ^j	0.68891 ± 0.00002 ^j	83.8058	0.68891 ± 0.00002
Kr	115.9 ± 0.2	0.7315 ± 0.0006	115.95	0.731	115.8	0.731	–	–	115.9 ± 0.2	0.7315 ± 0.0006
Xe	161.4 ± 0.3	0.816 ± 0.001	161.35	0.816	161.4	0.817	–	–	161.4 ± 0.3	0.816 ± 0.001

^a Triple point of H₂O reviewed by Wagner and Pruss (2002).

^b Triple point of O₂ measured by Mullins et al. (1963).

^c Triple point of O₃ measured by Hanson and Mauersberger (1986).

^d We obtained these triple point values by various extrapolations (see text).

^e Triple point of CH₄ reviewed by Friend et al. (1989).

^f Triple points of C₂H₆ and C₃H₈ reviewed by Younglove and Ely (1987).

^g Triple point of C₆H₆ reviewed by Goodwin (1988).

^h Triple point of N₂O measured by Yato et al. (1975).

ⁱ Triple point of AsH₃ measured by Sherman and Giauque (1955).

^j Triple point of Ar reviewed by Tegeler et al. (1999).

Table 3
Detailed bibliography.

Chemical name	Species	References	Temperature range (K)	Pressure range (bar)	Number of experimental points
Water	H ₂ O	Douslin and Osborn (1965)	241.72–271.09	3.32×10^{-4} – 5.16×10^{-3}	8
		Jancso et al. (1970)	194.64–271.68	8.00×10^{-7} – 5.40×10^{-3}	77
		Bryson et al. (1974)	131.80–187.02	8.44×10^{-13} – 1.97×10^{-10}	39
		Marti and Mauersberger (1993)	173.15–273.15	1.43×10^{-8} – 6.11×10^{-3}	11
		White et al. (1998)	130–167.2	4.78×10^{-13} – 3.21×10^{-9}	14
		Mauersberger and Krankowsky (2003)	164.5–169.9	1.64×10^{-9} – 7.07×10^{-9}	61
		Shilling et al.(2006)	181.3–191.2	4.2×10^{-8} – 2.48×10^{-7}	6
		All data	130–273.15	4.78×10^{-13} – 6.11×10^{-3}	216
Oxygen	O ₂	Hoge (1950)	51.30–53.09	6.27×10^{-4} – 1.01×10^{-3}	3
		Aoyama and Kanda (1934)	37.6–53	2.7×10^{-7} – 9.4×10^{-4}	10
		All data	37.6–53.09	2.7×10^{-7} – 1.01×10^{-3}	13
Ozone	O ₃	Hanson and Mauersberger (1986)	65.2–79.2	8.4×10^{-9} – 4.9×10^{-6}	25
Carbon monoxide	CO	Clayton and Giauque (1932)	60.24–67.02	2.664×10^{-2} –0.124	8
		Shinoda (1969)	54.78–68.07	6.050×10^{-3} –0.152	28
		All data	54.78–68.07	6.050×10^{-3} –0.152	36
Carbon dioxide	CO ₂	Ambrose (1956)	178.45–197.80	0.237–1.306	16
		Fernandez-Fassnacht and Rio (1984)	194.23–216.56	0.977–5.174	21
		Meyers and Van Dusen (1933)	194.35–216.52	0.990–5.171	28
		Giauque and Egan (1937)	154.20–195.83	1.511×10^{-2} –1.114	12
		Eiler et al. (2000)	130.1–169.6	3.333×10^{-4} – 9.354×10^{-2}	56
		Tickner and Lossing (1951)	106.45–153.65	1.333×10^{-6} – 1.333×10^{-2}	13
		Bryson et al. (1974)	69.69–102.5	1.15×10^{-13} – 4.213×10^{-7}	61
		All data	69.69–216.56	1.148×10^{-13} –5.174	207
Methanol	CH ₃ OH	Lucas et al. (2005)	133.4–173.7	8.6×10^{-11} – 1.3×10^{-6}	20
Formic acid	HCOOH	Coolidge (1930)	268.08–281.40	6.64×10^{-3} – 2.39×10^{-2}	3
Methane	CH ₄	Karwat (1924)	76.89–87.25	1.163×10^{-2} – 7.113×10^{-2}	8
		Tickner and Lossing (1951)	48.15–77.65	1.333×10^{-6} – 1.333×10^{-2}	13
		Armstrong et al. (1955)	53.15–90.66	1.073×10^{-5} –0.116	53
		All data	48.15–90.66	1.33×10^{-6} –0.116	74
Acetylene	C ₂ H ₂	McIntosh (1907)	177.85–191.95	0.405–1.263	9
		Tickner and Lossing (1951)	98.55–145.15	1.333×10^{-6} – 1.333×10^{-2}	13
		Burrell and Robertson (1915a)	129.9–189.1	1.33×10^{-3} –1.013	17
		All data	98.55–191.95	1.33×10^{-6} –1.263	39
Ethylene	C ₂ H ₄	Tickner and Lossing (1951)	79.45–102.45	1.33×10^{-6} – 6.66×10^{-4}	9
		Menaucourt (1982)	77.3–103.78	4.40×10^{-7} – 1.17×10^{-3}	16
		All data	77.3–103.78	4.40×10^{-7} – 1.17×10^{-3}	25
Ethane	C ₂ H ₆	Tickner and Lossing (1951)	83.35–88.75	1.33×10^{-6} – 6.66×10^{-6}	3
Propyne	C ₃ H ₄ (p)	Van Hook (1967)	160.7–168.2	1.73×10^{-3} – 3.85×10^{-3}	6
Butadiyne	C ₄ H ₂	Khanna et al. (1990)	127–152	6×10^{-10} – 2.5×10^{-8}	4
Hexatriyne	C ₆ H ₂	Shindo et al. (2003)	170.9–198.1	2.1×10^{-8} – 1.1×10^{-6}	6
Benzene	C ₆ H ₆	Deitz (1933)	184.3–200.2	1.93×10^{-6} – 1.04×10^{-5}	4
		Milazzo (1956)	195.18–273.07	8×10^{-6} – 3.26×10^{-2}	10
		Jackowski (1974)	220.79–278.68	3.20×10^{-4} – 4.78×10^{-2}	21
		Ha et al. (1976)	228.69–273.16	6.41×10^{-4} – 3.23×10^{-2}	14
		All data	184.3–278.68	1.93×10^{-6} – 4.78×10^{-2}	49
Hydrogen cyanide	HCN	Lewis and Schutz (1934)	236.2–259.3	3.6×10^{-2} –0.18	7
Propiolonitrile	HC ₃ N	Benilan et al. (1994)	164.9–201.5	1×10^{-6} – 1.2×10^{-4}	9
Cyanogen	C ₂ N ₂	Perry and Bardwell (1925)	179.94–240.79	2.3×10^{-3} –0.550	11
		Ruehrwein and Giauque (1939)	202.42–245.27	2.40×10^{-2} –0.738	10
		Benilan (1995)	156.3–169.5	5.2×10^{-5} – 4×10^{-4}	4
		All data	156.3–245.27	5.2×10^{-5} –0.738	25
Butynedinitrile	C ₄ N ₂	Saggiomo (1957)	262.6–273.2	2.15×10^{-2} – 4.72×10^{-2}	2
		Khanna et al. (1990)	147–152	3.6×10^{-10} – 7.6×10^{-9}	4
		Benilan (1995)	181.4–224	1.3×10^{-6} – 3.08×10^{-4}	10
		All data	147–273.2	3.6×10^{-10} – 4.72×10^{-2}	16
Nitrogen	N ₂	Giauque and Clayton (1933)	54.78–61.70	1.697×10^{-2} – 9.277×10^{-2}	8
		Borovik et al. (1960)	21.20–26.40	1.47×10^{-13} – 4.27×10^{-10}	9
		Frels et al. (1974)	35.40–59.17	4.840×10^{-6} – 5.113×10^{-2}	21
		All data	21.20–60.67	1.467×10^{-13} – 7.367×10^{-2}	38
Ammonia	NH ₃	Karwat (1924)	162.39–193.45	1.27×10^{-3} – 5.01×10^{-2}	8
		Overstreet and Giauque (1937)	176.92–195.36	8.2×10^{-3} – 6.08×10^{-2}	7
		All data	162.39–195.36	1.27×10^{-3} – 6.08×10^{-2}	15
Nitric oxide	NO	Johnston and Giauque (1929)	94.62–109.49	1.22×10^{-2} –0.219	8
		Ernest (1961)	64.85–94.95	6.66×10^{-7} – 1.33×10^{-2}	10

Table 3 (continued)

Chemical name	Species	References	Temperature range (K)	Pressure range (bar)	Number of experimental points
Nitrous oxide	N ₂ O	All data	64.85–109.49	6.66×10^{-7} –0.219	18
		Burrell and Robertson (1915b)	129.05–182.15	1.33×10^{-3} –0.867	14
		Black et al. (1930)	103.15–123.15	1.73×10^{-6} – 4.08×10^{-4}	7
		Blue and Giauque (1935)	148.48–182.26	2.43×10^{-2} –0.879	9
		Bryson et al. (1974)	68.1–80.2	1.14×10^{-12} – 1.21×10^{-9}	38
		Atake and Chihara (1974)	146.07–182.21	1.77×10^{-2} –0.865	26
		Yato et al. (1975)	143.88–182.29	1.31×10^{-2} –0.880	74
		Terlain (1983)	125.17–147.54	6.09×10^{-4} – 2.12×10^{-2}	12
Hydrogen sulphide	H ₂ S	All data	68.1–182.29	1.14×10^{-12} –0.88	180
		Giauque and Blue (1936)	164.90–182.84	3.21×10^{-2} –0.161	5
Sulphur dioxide	SO ₂	Burrell and Robertson (1915b)	178.75–197.15	6.66×10^{-4} – 1.6×10^{-2}	4
		Bergstrom (1922)	183.05–196.85	3.3×10^{-3} – 1.56×10^{-2}	4
		Schmitt and Rodriguez (2003)	120	1.38×10^{-8}	1
		All data	120–197.15	1.38×10^{-8} – 1.6×10^{-2}	9
Arsine	AsH ₃	Johnson and Pechukas (1937)	136.15–154.15	2.67×10^{-3} – 2.47×10^{-2}	13
Neon	Ne	Henning and Otto (1936)	22.04–23.64	0.131–0.288	11
		Bigeleisen and Roth (1961)	16.42–24.50	2.1×10^{-3} –0.418	91
		Ancsin (1978)	20.29–24.38	4.61×10^{-2} –0.400	34
		White et al. (1998)	5.98–9.13	1.39×10^{-14} – 8.27×10^{-9}	12
		Metcalf (2002)	5.50–7.00	4.80×10^{-16} – 4.62×10^{-12}	8
		All data	5.50–24.50	4.8×10^{-16} –0.418	156
Argon	Ar	Lee et al. (1970)	68.82–83.75	1.59×10^{-2} –0.683	133
		Leming and Pollack (1970)	35.51–83.73	3.06×10^{-9} –0.685	79
		Chen et al. (1978)	74.65–83.76	0.170–0.685	38
		White et al. (1998)	23.06–28.14	8.97×10^{-14} – 1.21×10^{-10}	5
		Metcalf (2002)	20.53–26.13	6.56×10^{-16} – 1.03×10^{-11}	13
		All data	20.53–83.76	6.56×10^{-16} –0.685	268
Krypton	Kr	Leming and Pollack (1970)	45.13–114.91	2.800×10^{-7} –0.674	106
		Fisher and McMillan (1958)	63.00–80.00	5.706×10^{-5} – 4.230×10^{-3}	18
		Chen et al. (1978)	103.72–115.63	0.192–0.720	37
		All data	45.13–115.63	2.800×10^{-7} –0.720	161
Xenon	Xe	Leming and Pollack (1970)	70.08–158.71	5.066×10^{-7} –0.675	118
		Chen et al. (1978)	142.83–161.33	0.183–0.813	60
		Tessier et al. (1982)	101.21–119.93	8.219×10^{-4} – 1.511×10^{-2}	12
		Bryson et al. (1974)	53.70–59.50	5.973×10^{-11} – 1.373×10^{-9}	7
		All data	53.70–161.33	5.973×10^{-11} –0.813	197

3.2.2. H₂O₂ (hydrogen peroxide)

The location of the triple point of H₂O₂ is unknown. Lide (2006) reports data only for the evaporation–liquefaction equilibrium and no measurements concerning the sublimation pressure have been found. Thus, no vapour pressure relation is proposed.

3.2.3. O₂ (oxygen)

The triple point of O₂ is located at 54.33 ± 0.06 K and $(1.49 \pm 0.03) \times 10^{-3}$ bar (Table 2). We have collected 13 experimental measurements ranging from 37.6 K (2.7×10^{-7} bar) to 53.09 K (1.01×10^{-3} bar; Aoyama and Kanda, 1934; Hoge, 1950; Table 3). Lide (2006) reports data only for the evaporation–liquefaction equilibrium. Solid O₂ has three different crystalline phases, the temperatures of the phase transitions are 23.781 K (α – β transition) and 43.772 K (β – γ transition; Mullins et al., 1963). Brown and Ziegler (1979) have computed three thermodynamic relations (Tables 4 and 5): the first one for $20 \text{ K} < T < 23.781 \text{ K}$ (α solid phase), the second one for $23.781 \text{ K} < T < 43.772 \text{ K}$ (β solid phase), and the last one for $43.772 \text{ K} < T < 54.33 \text{ K}$ (γ solid phase). The first relation cannot be checked as no experimental measurements have been performed for $T < 23.781 \text{ K}$. Nevertheless, the other two relations reproduce the measurements with an accuracy of $\pm 40\%$, which reflects the dispersion of the experimental points (Table 4 and Fig. 4). We consider that the thermodynamic relations calculated by Brown and Ziegler (1979)

are reliable and recommend their use. No other relations have been proposed in the astrophysical literature.

3.2.4. O₃ (ozone)

The determinations of the O₃ triple point location of the 'Air Liquide' company and of Hanson and Mauersberger (1986) do not agree (Table 2). Nevertheless, the data of Hanson and Mauersberger (1986) seem to be more realistic and we can consider that the triple point of O₃ is located at 79.6 ± 0.3 K and $(5.6 \pm 0.6) \times 10^{-6}$ bar (Table 2). We have collected 25 experimental measurements ranging from 65.2 K (8.4×10^{-9} bar) to 79.2 K (4.9×10^{-6} bar; Hanson and Mauersberger, 1986; Table 3 and Fig. 5). Lide (2006) reports data only for the evaporation–liquefaction equilibrium. No data concerning the heat capacity of the O₃ ice have been found. Thus we are not able to compute any thermodynamic relation. We recommend the use of the empirical relation calculated by Hanson and Mauersberger (1986), which reproduces their own experimental measurements with an accuracy of $\pm 3.7\%$ (Tables 4 and 5).

3.2.5. CO (carbon monoxide)

The triple point of CO is located at 68.10 ± 0.1 K and 0.1537 ± 0.0003 bar (Table 2). We have collected 36 experimental measurements ranging from 54.78 K (6.05×10^{-3} bar) to 68.07 K (0.152 bar; Clayton and Giauque, 1932; Shinoda, 1969; Table 3).

Table 4
 Informations on the polynomials of extrapolations.

Species	Polynomial designation	Solid phase	Type of extrapolation ^a	Temperature range (K) ^b	Number of data points	Degree of polynomial	Standard deviation (%)	Maximum deviation (%)	Reference
H ₂ O	H ₂ O-1	lh	T	0–273.16 (triple p.)	210	–	22	180	Feistel and Wagner (2007)
O ₂	O ₂ -1	α	T	20–23.78 (phase tr.)	0	3	–	–	Brown and Ziegler (1979)
	O ₂ -2	β	T	23.78 (phase tr.)–43.77 (phase tr.)	5	3	26	48	Brown and Ziegler (1979)
	O ₂ -3	γ	T	43.77 (phase tr.)–54.33 (triple p.)	8	3	19	43	Brown and Ziegler (1979)
O ₃	O ₃ -1	–	E	< 79.6 (triple p.)	25	1	2.3	4.1	Hanson and Mauersberger (1986)
CO	CO-1	α	T	14–61.55 (phase tr.)	14	5	1.2	3.2	This work
	CO-2	β	T	61.55 (phase tr.)–68.1 (triple p.)	22	3	0.9	2.0	Brown and Ziegler (1979)
CO ₂	CO ₂ -1	–	T	40–194.7	161	5	14.5	63	Brown and Ziegler (1979)
	CO ₂ -2	–	I	194.7–216.58 (triple p.)	46	2	0.33	0.72	This work
CH ₃ OH	CH ₃ OH-1	α	E	< 157.36 (phase tr.)	12	1	2.8	5.0	Lucas et al. (2005)
	CH ₃ OH-2	β	E	157.36 (phase tr.)–175.5 (triple pt)	8	1	2.8	5.7	Lucas et al. (2005)
HCOOH	HCOOH-1	–	E	< 281.4 (triple p.)	3	1	0.58	0.94	This work
	HCOOH-2	–	T	15–281.4 (triple p.)	3	4	0.9	1.2	This work
CH ₄	CH ₄ -1	I	T	20.6 (phase tr.)–90.68 (triple p.)	74	4	7.9	36	Brown and Ziegler (1979)
C ₂ H ₂	C ₂ H ₂ -1	–	E	< 192.3 (triple p.)	26	1	3.4	8.5	This work
C ₂ H ₄	C ₂ H ₄ -1	(II) & I	T	(20)–50–104.0 (triple p.)	25	4	9.3	29	Brown and Ziegler (1979)
C ₂ H ₆	C ₂ H ₆ -1	II & I	T	20–90.34 (triple p.)	3	5	16	22	Brown and Ziegler (1979)
C ₆ H ₆	C ₆ H ₆ -1	–	E	< 278.68 (triple p.)	49	1	11	41	This Work
HCN	HCN-1	–	T	15–259.85 (triple p.)	7	4	1.9	3.5	This work
HC ₃ N	HC ₃ N-1	–	E	< 201.5	9	1	3.9	7.1	This work
C ₂ N ₂	C ₂ N ₂ -1	–	E	< 245.32 (triple p.)	25	1	9.4	24	This work
C ₄ N ₂	C ₄ N ₂ -1	–	E	< 273.2	16	1	30	89	This work
N ₂	N ₂ -1	α	T	10–35.61 (phase tr.)	10	5	19	28	This work
	N ₂ -2	β	T	35.61 (phase tr.)–63.14 (triple p.)	28	4	1.5	4.6	This work
NH ₃	NH ₃ -1	–	T	15–195.41 (triple p.)	15	4	1.6	4.2	This work
NO	NO-1	–	E	< 109.50 (triple p.)	18	1	1.5	4.3	This work
	NO-2	–	I	64.85–109.50 (triple p.)	18	5	0.8	1.8	This work
N ₂ O	N ₂ O-1	–	E	< 182.30 (triple p.)	180	1	11	50	This work
	N ₂ O-2	–	I	68.1–182.30 (triple p.)	180	4	10	47	This work
H ₂ S	H ₂ S-1	–	E	< 187.57 (triple p.)	5	1	0.27	0.43	This Work
	H ₂ S-2	I	T	126.2 (phase tr.)–187.57 (triple p.)	5	4	0.03	0.05	This Work
SO ₂	SO ₂ -1	–	T	15–197.63 (triple p.)	8	4	5.0	9.6	This Work
AsH ₃	AsH ₃ -1	–	E	< 156.23 (triple p.)	13	1	2.6	6.0	This Work
Ne	Ne-1	–	T	3.7–12.5	20	3	94	158	Brown and Ziegler (1979)
	Ne-2	–	T	12.5–24.56 (triple p.)	136	4	2.8	8.6	Brown and Ziegler (1979)
Ar	Ar-1	–	T	20–83.79 (triple p.)	262	4	36	207	Brown and Ziegler (1979)
Kr	Kr-1	–	T	20–115.87 (triple p.)	145	6	7.1	35	Brown and Ziegler (1979)
Xe	Xe-1	–	T	20–161.4 (triple p.)	193	5	6.7	46	Brown and Ziegler (1979)

^a This column provides the type of extrapolation that is used. 'T', 'E' and 'I' stand for thermodynamic, empirical, and interpolation relations, respectively.

^b In this column, we have specified when the temperature limits correspond to a solid–solid phase transition (phase tr.) or to the triple point (triple p.). When nothing is specified, the temperature limit corresponds only to the limitation of the calculation.

Lide (2006) reports data for the sublimation and evaporation equilibria, which agree with the previous experimental measurements. CO exists in two different crystalline phases (α and β), the temperature of the phase transition is 61.55 ± 0.05 K (Clayton and Giauque, 1932). Brown and Ziegler (1979) have computed the thermodynamic relations of sublimation for both phases. At temperatures higher than 61.55 K (β phase), the interpolated thermodynamic relation proposed by Brown and

Ziegler (1979) reproduces the experimental data very well, with an accuracy of $\pm 2.4\%$ (Table 4). Thus, we did not compute a new thermodynamic relation but we give the interpolation coefficients from Brown and Ziegler (1979) in Table 5. For temperatures below 61.55 K (α phase), the relation proposed by Brown and Ziegler (1979) does not reproduce correctly the experimental measurements of Clayton and Giauque (1932) and Shinoda (1969). Therefore we propose our own thermodynamic

Table 5
Coefficients of the polynomials of extrapolations.

Species	Polynomial designation	A_0	A_1 (K)	A_2 (K ²)	A_3 (K ³)	A_4 (K ⁴)	A_5 (K ⁵)	A_6 (K ⁶)
H ₂ O	H ₂ O-1	See text as well as Eqs. (5) and (6) and Table 6						
O ₂	O ₂ -1	$1.541 \times 10^{+1}$	$-1.148 \times 10^{+3}$	$3.349 \times 10^{+2}$	$6.021 \times 10^{+1}$	0	0	0
	O ₂ -2	$1.335 \times 10^{+1}$	$-1.012 \times 10^{+3}$	$-2.971 \times 10^{+3}$	$2.926 \times 10^{+4}$	0	0	0
	O ₂ -3	$1.018 \times 10^{+1}$	$-8.035 \times 10^{+2}$	$-7.080 \times 10^{+3}$	$7.553 \times 10^{+4}$	0	0	0
O ₃	O ₃ -1	$1.746 \times 10^{+1}$	$-2.352 \times 10^{+3}$	0	0	0	0	0
CO	CO-1	$1.043 \times 10^{+1}$	$-7.213 \times 10^{+2}$	$-1.074 \times 10^{+4}$	$2.341 \times 10^{+5}$	$-2.392 \times 10^{+6}$	$9.478 \times 10^{+6}$	0
	CO-2	$1.025 \times 10^{+1}$	$-7.482 \times 10^{+2}$	$-5.843 \times 10^{+3}$	$3.939 \times 10^{+4}$	0	0	0
CO ₂	CO ₂ -1	$1.476 \times 10^{+1}$	$-2.571 \times 10^{+3}$	$-7.781 \times 10^{+4}$	$4.325 \times 10^{+6}$	$-1.207 \times 10^{+8}$	$1.350 \times 10^{+9}$	0
	CO ₂ -2	$1.861 \times 10^{+1}$	$-4.154 \times 10^{+3}$	$1.041 \times 10^{+5}$	0	0	0	0
CH ₃ OH	CH ₃ OH-1	$1.918 \times 10^{+1}$	$-5.648 \times 10^{+3}$	0	0	0	0	0
	CH ₃ OH-2	$1.706 \times 10^{+1}$	$-5.314 \times 10^{+3}$	0	0	0	0	0
HCOOH	HCOOH-1	$2.189 \times 10^{+1}$	$-7.213 \times 10^{+3}$	0	0	0	0	0
	HCOOH-2	$2.164 \times 10^{+1}$	$-6.942 \times 10^{+3}$	$-6.579 \times 10^{+4}$	$3.316 \times 10^{+6}$	$-6.004 \times 10^{+7}$	0	0
CH ₄	CH ₄ -1	$1.051 \times 10^{+1}$	$-1.110 \times 10^{+3}$	$-4.341 \times 10^{+3}$	$1.035 \times 10^{+5}$	$-7.910 \times 10^{+5}$	0	0
C ₂ H ₂	C ₂ H ₂ -1	$1.340 \times 10^{+1}$	$-2.536 \times 10^{+3}$	0	0	0	0	0
C ₂ H ₄	C ₂ H ₄ -1	$1.540 \times 10^{+1}$	$-2.206 \times 10^{+3}$	$-1.216 \times 10^{+4}$	$2.843 \times 10^{+5}$	$-2.203 \times 10^{+6}$	0	0
C ₂ H ₆	C ₂ H ₆ -1	$1.511 \times 10^{+1}$	$-2.207 \times 10^{+3}$	$-2.411 \times 10^{+4}$	$7.744 \times 10^{+5}$	$-1.161 \times 10^{+7}$	$6.763 \times 10^{+7}$	0
C ₆ H ₆	C ₆ H ₆ -1	$1.735 \times 10^{+1}$	$-5.663 \times 10^{+3}$	0	0	0	0	0
HCN	HCN-1	$1.393 \times 10^{+1}$	$-3.624 \times 10^{+3}$	$-1.325 \times 10^{+5}$	$6.314 \times 10^{+6}$	$-1.128 \times 10^{+8}$	0	0
HC ₃ N	HC ₃ N-1	$1.301 \times 10^{+1}$	$-4.426 \times 10^{+3}$	0	0	0	0	0
C ₂ N ₂	C ₂ N ₂ -1	$1.653 \times 10^{+1}$	$-4.109 \times 10^{+3}$	0	0	0	0	0
C ₄ N ₂	C ₄ N ₂ -1	$1.909 \times 10^{+1}$	$-6.036 \times 10^{+3}$	0	0	0	0	0
N ₂	N ₂ -1	$1.240 \times 10^{+1}$	$-8.074 \times 10^{+2}$	$-3.926 \times 10^{+3}$	$6.297 \times 10^{+4}$	$-4.633 \times 10^{+5}$	$1.325 \times 10^{+6}$	0
	N ₂ -2	8.514	$-4.584 \times 10^{+2}$	$-1.987 \times 10^{+4}$	$4.800 \times 10^{+5}$	$-4.524 \times 10^{+6}$	0	0
NH ₃	NH ₃ -1	$1.596 \times 10^{+1}$	$-3.537 \times 10^{+3}$	$-3.310 \times 10^{+4}$	$1.742 \times 10^{+6}$	$-2.995 \times 10^{+7}$	0	0
NO	NO-1	$1.691 \times 10^{+1}$	$-2.016 \times 10^{+3}$	0	0	0	0	0
	NO-2	$1.2352 \times 10^{+2}$	$-4.7607 \times 10^{+4}$	$7.7292 \times 10^{+6}$	$-6.4950 \times 10^{+8}$	$2.7061 \times 10^{+10}$	$-4.4739 \times 10^{+11}$	0
N ₂ O	N ₂ O-1	$1.622 \times 10^{+1}$	$-2.971 \times 10^{+3}$	0	0	0	0	0
	N ₂ O-2	6.5664	$1.2711 \times 10^{+3}$	$-6.6835 \times 10^{+5}$	$4.4959 \times 10^{+7}$	$-1.0967 \times 10^{+9}$	0	0
H ₂ S	H ₂ S-1	$1.298 \times 10^{+1}$	$-2.707 \times 10^{+3}$	0	0	0	0	0
	H ₂ S-2	8.933	$-7.260 \times 10^{+2}$	$-3.504 \times 10^{+5}$	$2.724 \times 10^{+7}$	$-8.582 \times 10^{+8}$	0	0
SO ₂	SO ₂ -1	$1.560 \times 10^{+1}$	$-3.508 \times 10^{+3}$	$-9.401 \times 10^{+4}$	$4.152 \times 10^{+6}$	$-6.946 \times 10^{+7}$	0	0
AsH ₃	AsH ₃ -1	$1.176 \times 10^{+1}$	$-2.382 \times 10^{+3}$	0	0	0	0	0
Ne	Ne-1	9.886	$-2.699 \times 10^{+2}$	$1.283 \times 10^{+2}$	$-1.624 \times 10^{+2}$	0	0	0
	Ne-2	$1.061 \times 10^{+1}$	$-3.086 \times 10^{+2}$	$9.860 \times 10^{+2}$	$-9.069 \times 10^{+3}$	$3.514 \times 10^{+4}$	0	0
Ar	Ar-1	$1.069 \times 10^{+1}$	$-8.932 \times 10^{+2}$	$-3.567 \times 10^{+3}$	$6.574 \times 10^{+4}$	$-4.280 \times 10^{+5}$	0	0
Kr	Kr-1	$1.077 \times 10^{+1}$	$-1.223 \times 10^{+3}$	$-8.903 \times 10^{+3}$	$2.635 \times 10^{+5}$	$-4.260 \times 10^{+6}$	$3.575 \times 10^{+7}$	$-1.210 \times 10^{+8}$
Xe	Xe-1	$1.098 \times 10^{+1}$	$-1.737 \times 10^{+3}$	$-1.332 \times 10^{+4}$	$4.349 \times 10^{+5}$	$-7.027 \times 10^{+6}$	$4.447 \times 10^{+7}$	0

The form of the polynomial is given in Eq. 4 and pressure is expressed in bar.

relation (Tables 4 and 5). This thermodynamic relation for the α phase has been calculated using the values of heat capacity of CO ices measured by Clayton and Giauque (1932) from 14 K to the temperature of the solid (α)–solid (β) transition and the constant value of $(7/2)R$ for the heat capacity of the gas, which corresponds to the value of a perfect diatomic gas. This relation reproduces the

experimental data with an accuracy of $\pm 3.3\%$, which reflects the dispersion of the experimental measurements. In most of the models of cometary activity (Priainik et al., 2004), the relation that is used for CO sublimation is the empirical one proposed by Fanale and Salvail (1990). This relation is erroneous as the triple point pressure is overestimated by about 10% and the slope

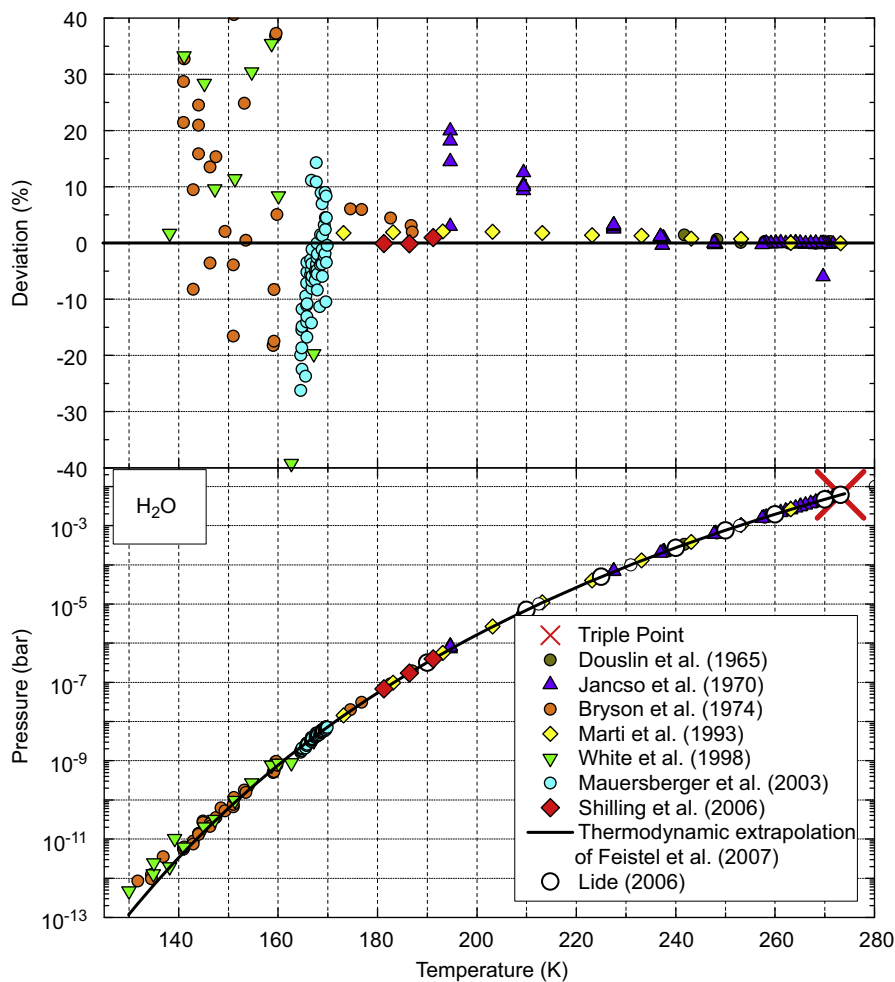


Fig. 3. Lower panel: sublimation vapour pressure of H₂O as a function of temperature. The experimental data are represented with different symbols, the triple point with a cross, and the thermodynamic relation of Feistel and Wagner (2007) with a black line. Upper panel: percentage deviation of the experimental data relative to the values calculated with the relation proposed by Feistel and Wagner (2007).

Table 6
Coefficients of the polynomial $\eta(T/T_i)$ (Eq. 6) of the semi-empirical relation (Eq. 5) for the sublimation pressure of crystalline Ih water ice (Feistel and Wagner, 2007).

<i>l</i>	<i>e_i</i>
0	20.996 966 510 789 7
1	3.724 374 782 713 62
2	−13.920 548 321 552 4
3	29.698 876 501 356 6
4	−40.197 239 263 594 4
5	29.788 048 105 021 5
6	−9.130 509 635 477 21

corresponds to that of the liquid–gas equilibrium (Fig. 6). In fact, this relation is an extrapolation of the liquid–gas equilibrium below the triple point temperature. In the astrophysical literature, two other empirical relations have been proposed by Yamamoto et al. (1983) and Huebner et al. (2006). The first does not reproduce correctly the experimental data (deviation of about 20% at 55 K) while the second relation proposed by Huebner et al. (2006) reproduces correctly the experimental data, but the slope below 55 K is not coherent with the measurements of heat capacity of the CO ices. Thus, we recommend the use of the thermodynamic relation proposed by Brown and Ziegler (1979) for the β phase ($T > 61.55$ K) and the one that we have computed for the α phase ($T < 61.55$ K; Tables 4 and 5).

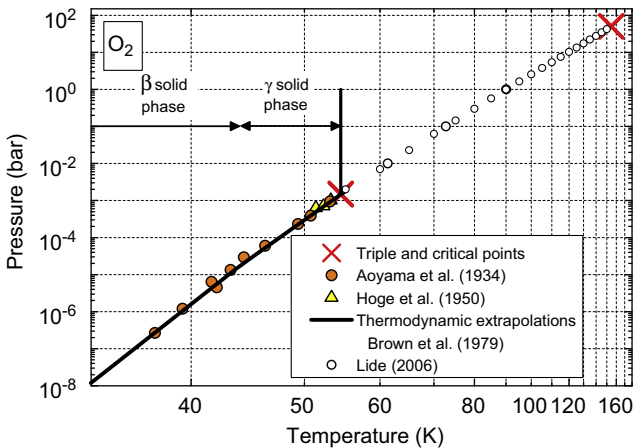


Fig. 4. Sublimation and evaporation pressures of O₂ as a function of temperature (reciprocal scale 1/*T*). Vertical lines show the temperatures of the β–γ phase transition and of the triple point.

3.2.6. CO₂ (carbon dioxide)

The triple point of CO₂ is located at 216.58 ± 0.03 K and 5.185 ± 0.005 bar (Table 2). We have collected 207 experimental measurements ranging from 69.69 K (1.148×10^{-13} bar) to 216.56 K

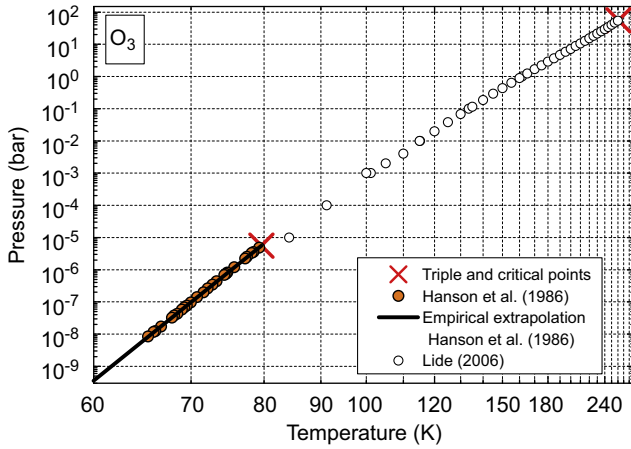


Fig. 5. Sublimation and evaporation pressures of O_3 as a function of temperature (reciprocal scale $1/T$).

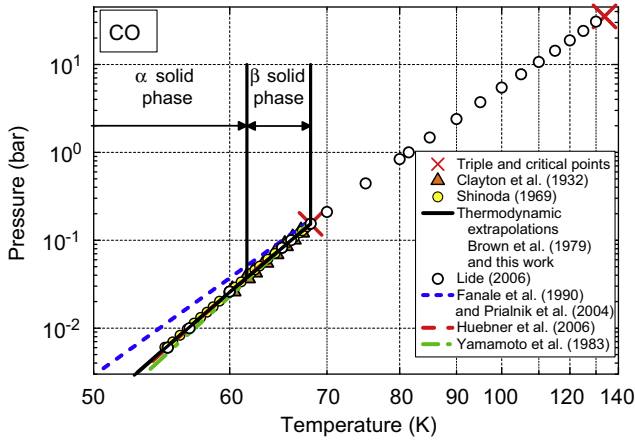


Fig. 6. Sublimation and evaporation pressures of CO as a function of temperature (reciprocal scale $1/T$). Vertical lines show the temperatures of the β - γ phase transition and of the triple point.

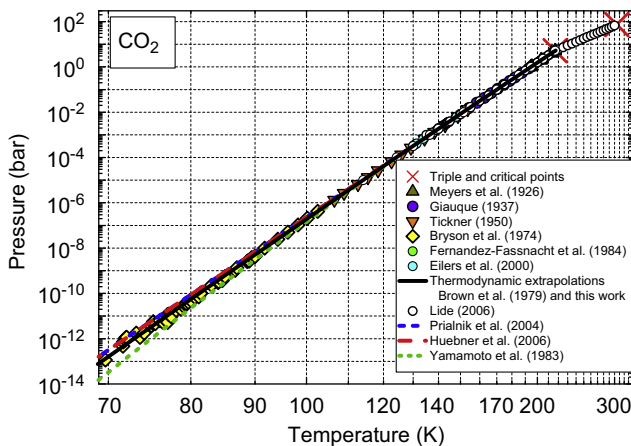


Fig. 7. Sublimation and evaporation pressures of CO_2 as a function of temperature (reciprocal scale $1/T$).

(5.174 bar; Meyers and Van Dusen, 1933; Giauque and Egan, 1937; Tickner and Lossing, 1951; Ambrose, 1956; Bryson et al., 1974; Fernandez-Fassnacht and Rio, 1984; Eiler et al., 2000; Table 3). Lide (2006) reports data for sublimation and evaporation equilibria, which agree with previous experimental measurements. Brown and Ziegler (1979) have proposed a thermodynamic extrapolation for

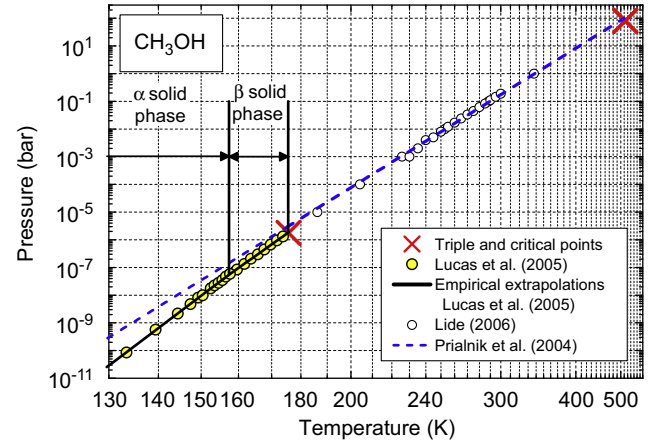


Fig. 8. Sublimation and evaporation pressures of CH_3OH as a function of temperature (reciprocal scale $1/T$). Vertical lines show the temperatures of the α - β phase transition and of the triple point.

temperatures ranging from 40 to 194.7 K (Tables 4 and 5), which fit with experimental measurements with an accuracy of $\pm 63\%$, reflecting the dispersion of the experimental data at $T < 120$ K. For $120 \text{ K} < T < 194.7 \text{ K}$, the accuracy of this relation is about $\pm 9\%$. For $194.7 \text{ K} < T < 216.58 \text{ K}$, we have performed an empirical interpolation of experimental data (Tables 4 and 5). Empirical extrapolation relations given by Yamamoto et al. (1983), Prialnik et al. (2004), and Huebner et al. (2006) reproduce very well experimental data for temperature higher than 120 K. Nevertheless, at lower temperature, Yamamoto et al. (1983) and Prialnik et al. (2004) overestimate the sublimation pressure (by a factor of ~ 2 at 80 K) whereas Huebner et al. (2006) underestimate it (factor ~ 0.4 at 80 K; Fig. 7). Thus, we recommend the use of the thermodynamic relation proposed by Brown and Ziegler (1979) below 194.7 K and our empirical relation above this temperature (Tables 4 and 5).

3.2.7. H_2CO (formaldehyde)

The triple point of H_2CO is located at $155.1 \pm 0.3 \text{ K}$, but its pressure is unknown (Table 2). From the data of Spence and Wild (1935), concerning the evaporation equilibrium, we can estimate that the pressure at the triple point is about $(4 \pm 1) \times 10^{-4} \text{ bar}$. Lide (2006) reports data only for the evaporation equilibrium. No experimental data concerning the sublimation equilibrium have been found. Thus, no extrapolation is proposed here.

3.2.8. CH_3OH (methanol)

The triple point of CH_3OH is located at $175.5 \pm 0.5 \text{ K}$ (Table 2). The pressure at the triple point has not been directly measured. Nevertheless, thanks to the data of Lucas et al. (2005), we can estimate that it should be about $(1.8 \pm 0.2) \times 10^{-6} \text{ bar}$. We have collected 20 experimental measurements ranging from 133.4 K ($8.6 \times 10^{-11} \text{ bar}$) to 173.7 K ($1.3 \times 10^{-6} \text{ bar}$; Lucas et al., 2005; Table 3). Lide (2006) reports data only for the evaporation equilibrium. CH_3OH exists in two different crystalline phases (α and β); the temperature of the phase transition is 157.36 K (Lucas et al., 2005; Carlson and Westrum, 1971). Below this temperature, CH_3OH can also exist in a metastable form (Lucas et al., 2005), for which the sublimation pressure is slightly higher than for the stable phase. The heat capacity of methanol ice has been measured by Carlson and Westrum (1971) and Kelley (1929). Nevertheless, as the internal partition function of gaseous CH_3OH is not reported in the HITRAN database, we cannot calculate the heat capacity of the gas and consequently cannot propose any thermodynamic relation. In Tables 4 and 5, we give the coefficients of empirical extrapolation relations determined for

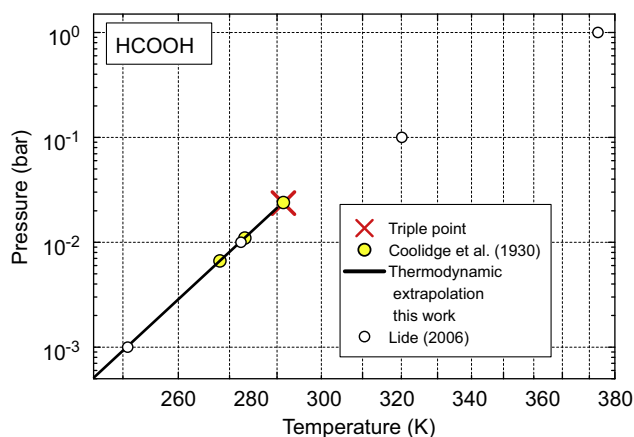


Fig. 9. Sublimation and evaporation pressures of HCOOH as a function of temperature (reciprocal scale $1/T$).

both crystalline phases by Lucas et al. (2005) from their own measurements. We can see in Fig. 8 that the empirical extrapolation proposed by Prialnik et al. (2004) is wrong as it is a direct extrapolation of the evaporation equilibrium curve for temperatures below the triple point, which is overestimated by a factor 1.7. Thus, we recommend the use of the empirical relations of Lucas et al. (2005) (Tables 4 and 5).

3.2.9. HCOOH (formic acid)

The triple point of HCOOH is located at 281.42 ± 0.07 K and 0.0236 ± 0.0001 bar (Table 2). We have collected only 3 experimental measurements ranging from 268.08 K (6.64×10^{-3} bar) to 281.40 K (2.39×10^{-2} bar; Coolidge, 1930; Table 3). Lide (2006) reports data for the sublimation and evaporation equilibria, which agree with the previous experimental measurements. No extrapolation relation of HCOOH vapour pressure as a function of temperature has been proposed in the literature (Table 1). Nevertheless, the heat capacity of the solid has been measured from 15 K to the triple point by Stout and Fisher (1941) and the heat capacity of the gas can be calculated down to 15 K thanks to the internal partition function given in the HITRAN database (Rothman et al., 2005), which has been extrapolated using an $T^{1.5}$ dependence for $T < 70$ K. We have computed a thermodynamic relation from 15 K to the triple point temperature (Tables 4 and 5 and Fig. 9). This relation reproduces the 3 experimental measurements with an accuracy of $\pm 1.2\%$. However, due to their limited temperature range the validation of this relation may still be questionable. Nevertheless, we predict a sublimation enthalpy of 61.2 kJ mol^{-1} at 213.15 K, which is very close to the value of $62.1 \pm 1.2 \text{ kJ mol}^{-1}$ measured by Calis-Van Ginkel et al. (1978) at this temperature. Thus, our thermodynamic relation seems to be accurate. We have also performed an empirical extrapolation based on the 3 experimental data (Tables 4 and 5). The thermodynamic relation differs from this extrapolation by only 18% at 200 K and 66% at 150 K. We recommend the use of this thermodynamic relation but new experimental measurements are required to check the validity at low temperature.

3.2.10. HCOOCH₃ (methyl formate)

The location of the triple point of HCOOCH₃ is unknown. Lide (2006) reports data only for the evaporation–liquefaction equilibrium and no measurements concerning the sublimation pressure have been found. Thus, no vapour pressure relation is proposed.

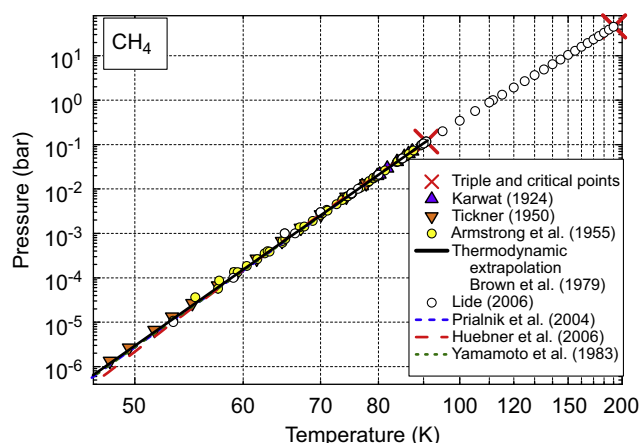


Fig. 10. Sublimation and evaporation pressures of CH₄ as a function of temperature (reciprocal scale $1/T$).

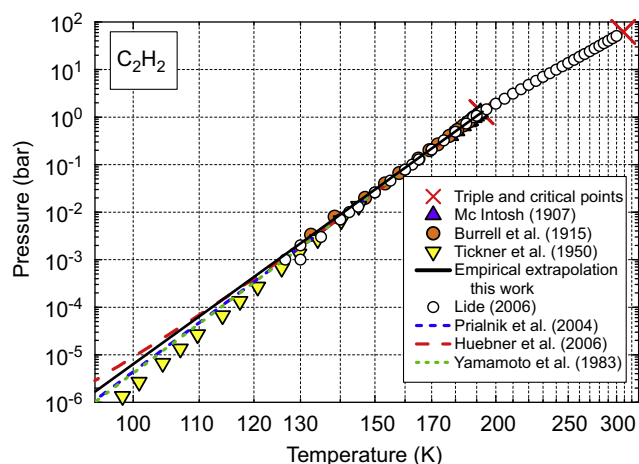


Fig. 11. Sublimation and evaporation pressures of C₂H₂ as a function of temperature (reciprocal scale $1/T$).

3.2.11. HOCH₂CH₂OH (ethylene glycol)

The temperature of the HOCH₂CH₂OH triple point is 258.8 ± 2.4 K and the pressure is unknown (Table 2). Lide (2006) reports data only for the evaporation equilibrium. No data concerning the sublimation equilibrium have been found.

3.2.12. CH₄ (methane)

Friend et al. (1989) have very accurately determined the location of the triple point of CH₄ ($T_t = 90.6854 \pm 0.0003$ K and $P_t = 0.11696 \pm 0.00002$ bar). We have collected 74 experimental measurements ranging from 48.15 K (1.33×10^{-6} bar) to 90.66 K (0.116 bar; Karwat, 1924; Tickner and Lossing, 1951; Armstrong et al., 1955; Table 3 and Fig. 10). Lide (2006) reports data for the sublimation and evaporation equilibria, which agree with the previous experimental measurements. Vogt and Pitzer (1976) have shown that CH₄ can exist in two different crystalline forms (phases II and I); the transition between both forms occurs at 20.53 K. Brown and Ziegler (1979) have calculated the thermodynamic relation for phase I of CH₄ ice between 20.6 and 90.68 K. The experimental measurements are reproduced with an accuracy of $\pm 40\%$, reflecting the dispersion of experimental data at $T < 70$ K. For $70 \text{ K} < T < 90.68 \text{ K}$, the uncertainty of this relation is about $\pm 3.5\%$. All empirical relations published in astrophysical literature by Yamamoto et al. (1983), Prialnik et al. (2004), and

Huebner et al. (2006) present higher deviations and therefore we recommend the use of the thermodynamic relation calculated by Brown and Ziegler (1979) (Tables 4 and 5).

3.2.13. C_2H_2 (acetylene)

The triple point of C_2H_2 is located at 192 ± 1 K and 1.275 ± 0.015 bar (Table 2). We have collected 39 experimental measurements ranging from 98.55 K (1.33×10^{-6} bar) to 191.95 K (1.263 bar; McIntosh, 1907; Burrell and Robertson, 1915a; Tickner and Lossing, 1951) (Table 3). We have excluded the measurements of Masterson et al. (1990), which differ significantly from all the other data. Moreover, close to 130 K, we can observe a discrepancy of about 40% between the data of Burrell and Robertson (1915a) and those of Tickner and Lossing (1951) (Fig. 11). As the measurements of Tickner and Lossing (1951) on C_2H_4 and C_2H_6 are also doubtful, we consider only the data of McIntosh (1907) and Burrell and Robertson (1915a). The different datasets which are reported by Lide (2006) are not in agreement (Fig. 11). No data concerning the heat capacity of the ice have been found and therefore we propose an empirical extrapolation relation (Tables 4 and 5). The empirical extrapolation relations proposed by Yamamoto et al. (1983), Prialnik et al. (2004), and Huebner et al. (2006) present higher deviations than our calculation. Hence we recommend our empirical relation (Tables 4 and 5). Further experimental studies are required to determine with more precision the sublimation pressure at temperatures lower than 180 K.

3.2.14. C_2H_4 (ethylene)

The triple point of C_2H_4 is located at 104.0 ± 0.1 K and $(1.2 \pm 0.1) \times 10^{-3}$ bar (Table 2). We have collected 25 experimental measurements ranging from 77.3 K (4.4×10^{-7} bar) to 103.78 K (1.17×10^{-3} bar; Tickner and Lossing, 1951; Menaucourt, 1982; Table 3 and Fig. 12). Lide (2006) reports data only for evaporation equilibrium. Brown and Ziegler (1979) have proposed a thermodynamic relation for temperatures ranging from 20 to 103.97 K (Tables 4 and 5). No other extrapolation relation has been proposed. The slopes of the curve related to the experimental data points of Tickner and Lossing (1951) and Menaucourt (1982) are different (Fig. 12). The measurements of Menaucourt (1982) agree with the calculation of Brown and Ziegler (1979) within $\pm 4\%$, which reflects the dispersion of these data. Thus, it seems that the sublimation relation proposed by Brown and Ziegler (1979) is reliable, whereas the data of Tickner and Lossing (1951), as for C_2H_2 and C_2H_6 , are inaccurate. Here, we recommend the thermodynamic relation computed by Brown and Ziegler (1979).

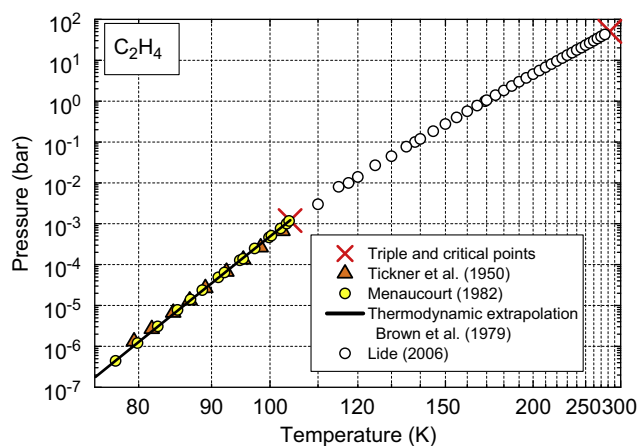


Fig. 12. Sublimation and evaporation pressures of C_2H_4 as a function of temperature (reciprocal scale $1/T$). A phase transition occurs near 50 K.

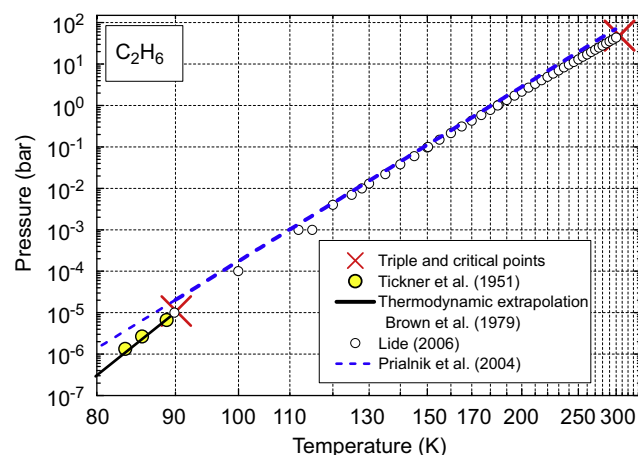


Fig. 13. Sublimation and evaporation pressures of C_2H_6 as a function of temperature (reciprocal scale $1/T$).

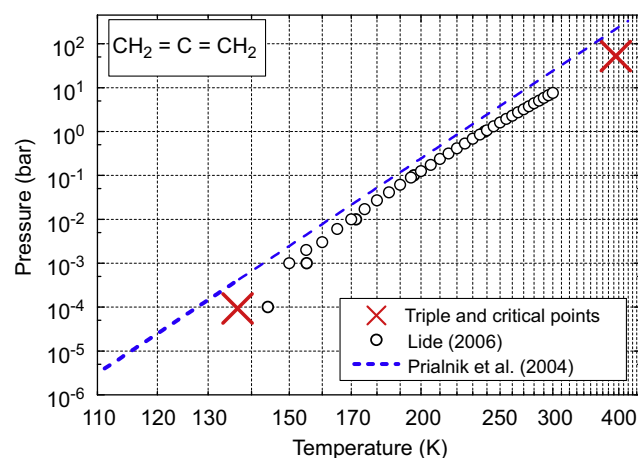


Fig. 14. Evaporation pressure of $CH_2=C=CH_2$ as a function of temperature (reciprocal scale $1/T$).

However, as there is a phase change occurring around 50 K (transition II–I; Rytter and Gruen, 1979), not taken into account in the calculations of Brown and Ziegler (1979), the validity of this relation may diverge below this temperature.

3.2.15. C_2H_6 (ethane)

The triple point of C_2H_6 is located at 90.34 ± 0.01 K and $(1.1 \pm 0.1) \times 10^{-5}$ bar (Table 2). We have collected only 3 experimental measurements ranging from 83.35 K (1.3×10^{-6} bar) to 88.75 K (6.7×10^{-6} bar; Tickner and Lossing, 1951; Table 3). Lide (2006) reports only data for evaporation equilibrium. Therefore, the sublimation properties of C_2H_6 are poorly known. C_2H_6 ice can exist in two different crystalline forms (phases II and I), and the transition temperature (89.77 ± 0.13 K) is very close to the triple point (Hans, 1976). Moreover, a metastable solid phase may also exist at lower temperatures (Wisnosky et al., 1983). Brown and Ziegler (1979) have computed a thermodynamic relation for temperatures ranging from 20 to 89.9 K (Tables 4 and 5), which does not reproduce correctly the experimental data of Tickner and Lossing (1951) (Fig. 13). However, as we saw earlier, in the cases of C_2H_2 and C_2H_4 , the measurements of Tickner and Lossing (1951) do not seem accurate. Thus, we are rather confident in the relation proposed by Brown and Ziegler (1979), but experimental measurements are required to validate this relation. Prialnik et al. (2004) have proposed an empirical sublimation relation for C_2H_6 , which is false. Indeed, at the triple point temperature, it

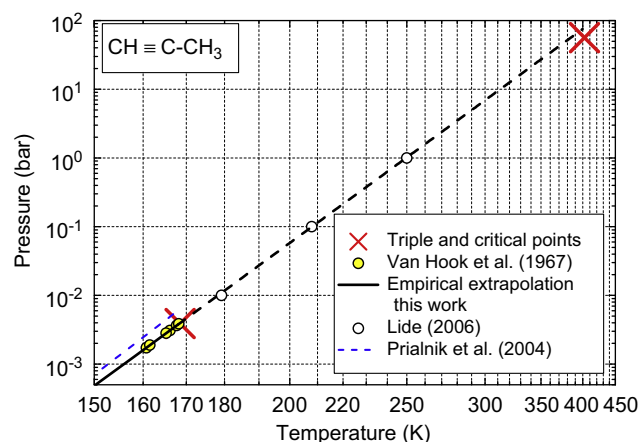


Fig. 15. Evaporation pressure of $\text{CH}_3\text{-C}\equiv\text{CH}$ as a function of temperature (reciprocal scale $1/T$). The dashed black line represents the empirical relation that we have calculated from the experimental data of Van Hook (1967) for $T < 168.5$ K, which is extrapolated for $T > 168.5$ K.

overestimates the sublimation pressure by a factor 2 and the slope of this curve is approximately the same as the evaporation equilibrium curve (Fig. 13). Thus, we recommend the use of the thermodynamic relation computed by Brown and Ziegler (1979).

3.2.16. C_3H_4 ($\text{CH}_2=\text{C}=\text{CH}_2$) (allene)

The temperature at the triple point of $\text{CH}_2=\text{C}=\text{CH}_2$ is 136.59 ± 0.20 K and its pressure is unknown (Table 2). Nevertheless, by extrapolation of the evaporation data of Stull (1947a), we can estimate that the pressure at the triple point of $\text{CH}_2=\text{C}=\text{CH}_2$ is about $9.3 \pm 0.4 \times 10^{-5}$ bar. Lide (2006) reports data only for the evaporation equilibrium (Fig. 14) and no experimental data concerning sublimation have been found. Nevertheless, Prialnik et al. (2004) have proposed an empirical relation for the sublimation pressure, which has a slope close to that of the vaporization equilibrium curve (Fig. 14) and which overestimates the data of Lide (2006) by a factor of approximately 3. Thus, this relation is false and should not be used. Unfortunately, no other sublimation relation could be proposed as the sublimation vapour pressure of $\text{CH}_2=\text{C}=\text{CH}_2$ has never been measured, nor has the heat capacity of solid allene been measured.

3.2.17. C_3H_4 ($\text{CH}_3\text{-C}\equiv\text{CH}$) (propyne)

The triple point of $\text{CH}_3\text{-C}\equiv\text{CH}$ is located at 168.5 ± 0.3 K and its pressure is unknown (Table 2). From the data of Van Hook (1967), we can estimate that the pressure at the triple point should be about $3.9 \pm 0.2 \times 10^{-3}$ bar. We have collected only 6 experimental measurements ranging from 160.7 K (1.73×10^{-3} bar) to 168.2 K (3.85×10^{-3} bar; Van Hook, 1967; Table 3). Lide (2006) reports data only for the evaporation–liquefaction equilibrium (Fig. 15). We have performed an empirical extrapolation of the sublimation data of Van Hook (1967), which reproduces also the data of the evaporation equilibrium (Fig. 15). So, we think that these measurements of Van Hook (1967) probably correspond to the vapour pressure of supercooling liquid propyne and not of ice. Thus, no measurement of the sublimation pressure of propyne is available. As for allene, Prialnik et al. (2004) have proposed an empirical relation for the sublimation pressure, which has a slope similar to that of the vaporization equilibrium but gives values higher by a factor of approximately 2 compared with the data of Van Hook (1967). Therefore, this sublimation relation is false and should not be used. Unfortunately, as for $\text{CH}_2=\text{C}=\text{CH}_2$, no other sublimation relation can be proposed as the vapour pressure of $\text{CH}_3\text{-C}\equiv\text{CH}$ ice has never been reported.

3.2.18. C_3H_6 (propene)

The triple point of C_3H_6 is located at low temperature (87.8 ± 0.8 K) and its pressure has never been measured (Table 2). Nevertheless, the extrapolation of the evaporation data of Powell and Giauque (1939) and Stull (1947a) allows estimation of the C_3H_6 triple point pressure between 10^{-7} and 10^{-8} bar. Lide (2006) reports data only for the evaporation equilibrium and no data concerning the sublimation have been found. Thus no sublimation relation can be proposed.

3.2.19. C_3H_8 (propane)

The triple point of C_3H_8 is located at low temperature (85.47 ± 0.05 K) and at extremely low pressure ($(1.685 \pm 0.001) \times 10^{-9}$ bar; Younglove and Ely, 1987; Table 2). Lide (2006) reports data only for the evaporation equilibrium and no data concerning the sublimation have been found. Thus no sublimation relation can be proposed.

3.2.20. C_4H_2 (diacetylene)

The location of the triple point of C_4H_2 is not precisely known. According to Stull (1947a), its temperature and pressure are 235.6 ± 3.5 K and 0.1 ± 0.03 bar, respectively (Table 2). Khanna et al. (1990) have measured the vapour pressures of ice between 127 and 152 K (Table 3). These authors point out that their measurements are reliable only within a factor 2.5–3. Thus, these measurements seem doubtful, and we propose no empirical sublimation relation. Moreover Lide (2006) reports no data concerning the vapour pressure of C_4H_2 . Thus, new experimental measurements are required to determine the vapour pressure of C_4H_2 .

3.2.21. C_6H_2 (triacetylene)

The location of the triple point of C_6H_2 is unknown. Shindo et al. (2003) have measured the vapour pressure between 170.8 K (2.1×10^{-8} bar) and 198.1 K (1.1×10^{-6} bar) (Table 3). Nevertheless, it is not clear if their measurements correspond to the vaporization equilibrium or to the sublimation equilibrium. No other data have been published elsewhere. Thus no sublimation relation can be proposed.

3.2.22. C_6H_6 (benzene)

The triple point of C_6H_6 is located at 278.68 ± 0.02 K and $(4.785 \pm 0.005) \times 10^{-2}$ bar (Table 2; Goodwin, 1988). We have collected 49 experimental measurements ranging from 184.3 K (1.93×10^{-6} bar) to 278.68 K (4.78×10^{-2} bar; Deitz, 1933;

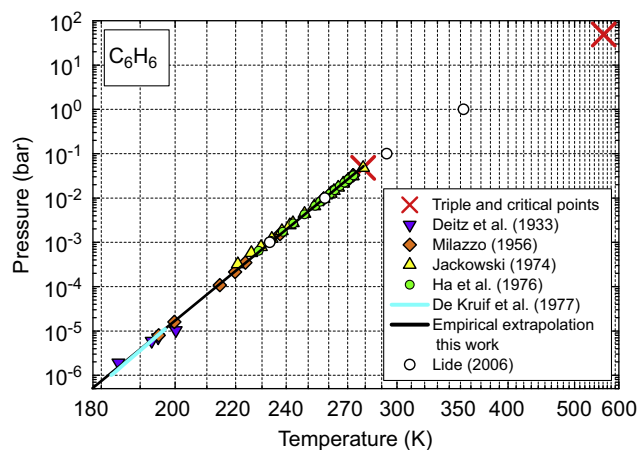


Fig. 16. Sublimation and evaporation pressures of C_6H_6 as a function of temperature (reciprocal scale $1/T$).

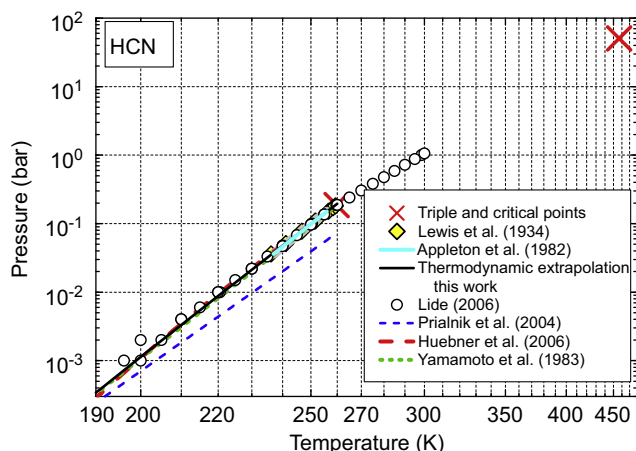


Fig. 17. Sublimation and evaporation pressures of HCN as a function of temperature (reciprocal scale $1/T$).

Milazzo, 1956; Jackowski, 1974; Ha et al., 1976; Table 3). De Kruif and Van Ginkel (1977) have also measured the vapour pressure of solid C_6H_6 between 183 and 197 K, but unfortunately their results are not tabulated; they give only an empirical relation of the form $\ln(P)=A+B/T$. As can be seen, in Fig. 16, this empirical relation and the data reported by Lide (2006) are in agreement with the other experimental measurements. So far, no other sublimation relation for C_6H_6 has been proposed. The heat capacity of solid C_6H_6 has been measured by Ziegler and Andrews (1942). Nevertheless, as the internal partition function of gaseous C_6H_6 is not reported in the HITRAN database, we cannot calculate the heat capacity of the gas and therefore cannot propose any thermodynamic relation. We can only propose an empirical extrapolation relation (Tables 4 and 5), which reproduces the experimental data with an accuracy of $\pm 41\%$, reflecting the dispersion of the measurements. We recommend the use of this empirical relation.

3.2.23. HCN (hydrogen cyanide)

The triple point of HCN is located at 259.85 ± 0.03 K and 0.1866 ± 0.0004 bar (Table 2). We have collected only 7 experimental measurements ranging from 235.2 K (3.6×10^{-2} bar) to 259.3 K (0.18 bar; Lewis and Schutz, 1934; Table 3). Appleton and Van Hook (1982) have also measured the vapour pressure of solid HCN between 237 and 256 K. Unfortunately their results are not tabulated. They give only an empirical relation of the form $\ln(P)=A+B/T$. Nevertheless, their results are in agreement with the measurements of Lewis and Schutz (1934) (Fig. 17). We can note in Fig. 17 that the different datasets reported by Lide (2006) are not in agreement around 200 K. The heat capacity of the solid has been measured from 15 K to the triple point by Giauque and Ruehrwein (1939) and the heat capacity of the gas can be calculated down to 15 K thanks to the internal partition function given in the HITRAN database (Rothman et al., 2005), which has been extrapolated using a linear dependence with temperature for $T < 70$ K. From these data, we have computed a thermodynamic relation from 15 K to the triple point temperature (Tables 4 and 5). This relation reproduces the experimental data very well, with an accuracy of $\pm 3.5\%$, which reflects the dispersion of the measurements. Some other empirical relations have been proposed by Yamamoto et al. (1983), Prialnik et al. (2004), and Huebner et al. (2006). The relation proposed by Huebner et al. (2006) reproduces the experimental data with an accuracy of $\pm 3\%$. Nevertheless, its slope at $T < 180$ K is not in agreement with the values of the heat capacities. The slope of the relation of Yamamoto et al. (1983) is too large and thus reproduces the

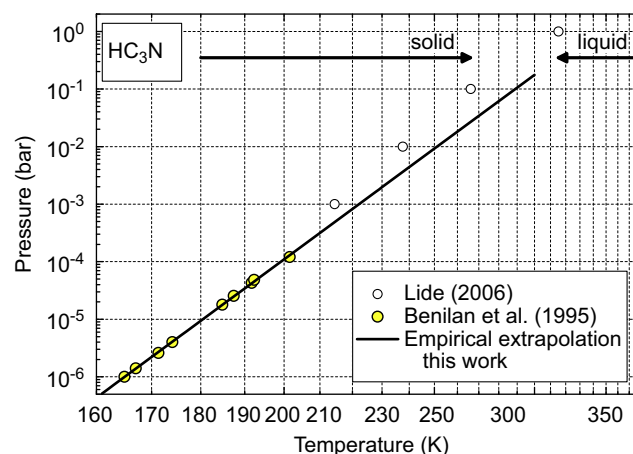


Fig. 18. Sublimation and evaporation pressures of HC_3N as a function of temperature (reciprocal scale $1/T$).

experimental data with an accuracy of only $\pm 10\%$. The empirical relation proposed by Prialnik et al. (2004) underestimates all the measurements, including the triple point, by almost a factor of 2 (Fig. 17). Both relations should not be used, and we rather recommend the thermodynamic relation that we have computed.

3.2.24. HNC (hydrogen isocyanide)

The location of the triple point of HNC is unknown. No measurements concerning the vapour pressure of HNC have been found. Thus, no vapour pressure relation is proposed.

3.2.25. HC_3N (propiolonitrile)

It seems that the location of the triple point has never been determined directly. Nevertheless, Lide (2006) reports that at 266 K ($P_{vap}=0.1$ bar) HC_3N is solid, whereas at 315 K ($P_{vap}=1$ bar) it is liquid (Fig. 18). Thus, the triple point should be located between these two points. We have collected 9 experimental measurements ranging from 164.9 K (10^{-6} bar) to 201.5 K (1.2×10^{-4} bar; Benilan et al., 1994; Benilan, 1995; Table 3), which are not in agreement with Lide (2006). Nevertheless, we propose an empirical extrapolation relation (see Tables 4 and 5) based on the measurements of Benilan (1995), reproducing the experimental data with an accuracy of $\pm 8\%$, which reflects the dispersion of the measurements. As no other relation has been proposed, we recommend the use of this empirical relation. Further experimental studies are required to determine the sublimation pressure of HC_3N with more accuracy.

3.2.26. CH_3CN (acetonitrile)

The triple point of CH_3CN is located at low temperature, 229.32 ± 0.02 K, and its pressure is unknown (Table 2). Thanks to empirical extrapolation of the evaporation data of Stull (1947a), we can estimate that the pressure at the triple point is about $(2.0 \pm 0.1) \times 10^{-3}$ bar. Lide (2006) reports data only for the evaporation equilibrium. It has been shown that CH_3CN ice exists in two different crystalline form (phases II and I) with a transition temperature of 216.9 ± 0.1 K (Putnam et al., 1965). Nevertheless, no data concerning its sublimation have been found. Thus no sublimation relation can be proposed.

3.2.27. CH_3CH_2CN (propanenitrile)

The triple point of CH_3CH_2CN is located at low temperature, 180.37 ± 0.02 K, and its pressure is unknown (Table 2). Thanks to empirical extrapolation of the evaporation data of Stull (1947a), we can estimate that the pressure at the triple point is about

$(3.8 \pm 0.1) \times 10^{-6}$ bar. Lide (2006) reports data only for the evaporation equilibrium. It has been shown that $\text{CH}_3\text{CH}_2\text{CN}$ ice exists in two different crystalline forms (phases II and I) with a transition temperature of 174.96 ± 0.01 K (Weber and Kilpatrick, 1962). Nevertheless, no data on the sublimation equilibrium have been found and no sublimation relation can be proposed.

3.2.28. C_2N_2 (cyanogen)

The triple point of C_2N_2 is located at 245.32 ± 0.02 K and 0.744 ± 0.009 bar (Table 2). We have collected 25 experimental measurements ranging from 156.3 K (5.2×10^{-5} bar) to 245.27 K (0.738 bar; Perry and Bardwell, 1925; Ruehrwein and Giauque, 1939; Benilan, 1995; Table 3). Lide (2006) reports data for the sublimation and evaporation equilibria, which agree with the previous experimental measurements. The heat capacity of solid C_2N_2 has been measured by Ruehrwein and Giauque (1939). Nevertheless, as the internal partition function of gaseous C_2N_2 is not reported in the HITRAN database, we cannot calculate the heat capacity of the gas and so no thermodynamic relation is proposed. Therefore, we propose an empirical relation (Tables 4 and 5), reproducing the experimental data with an accuracy of $\pm 24\%$, which reflects the dispersion of the measurements (Fig. 19). The relation proposed by Huebner et al. (2006) reproduces also the experimental data with an accuracy of

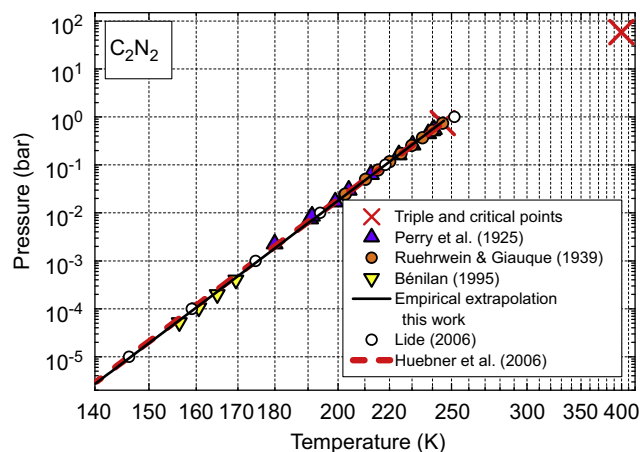


Fig. 19. Sublimation and evaporation pressures of C_2N_2 as a function of temperature (reciprocal scale $1/T$).

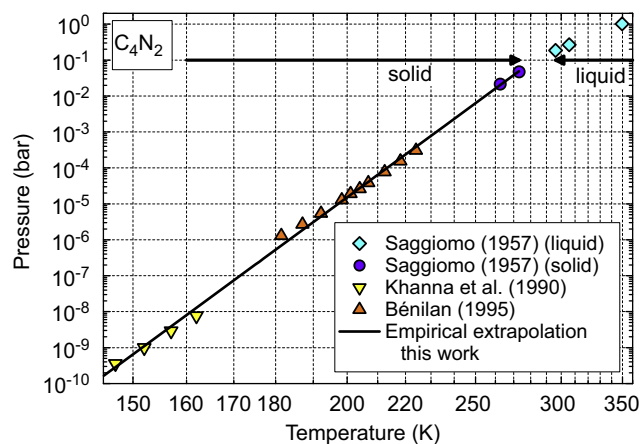


Fig. 20. Sublimation and evaporation pressures of C_4N_2 as a function of temperature (reciprocal scale $1/T$).

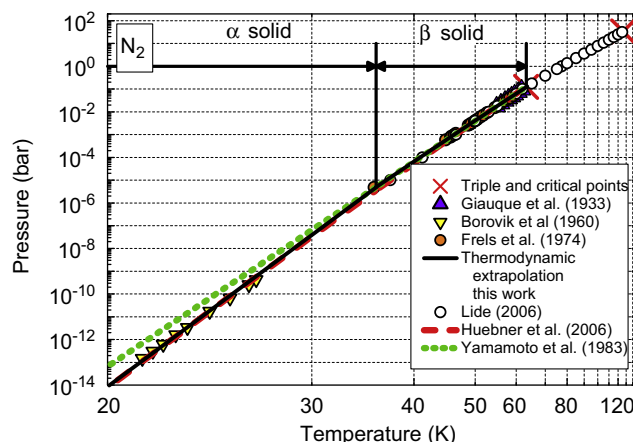


Fig. 21. Sublimation and evaporation pressures of N_2 as a function of temperature (reciprocal scale $1/T$). Vertical lines show the temperatures of the α - β phase transition and of the triple point.

$\pm 20\%$. Nevertheless, its slope for $T < 120$ K seems to be too high. Thus, we recommend the use of our empirical relation.

3.2.29. C_4N_2 (butynedinitrile)

It seems that the location of the triple point of C_4N_2 has never been determined directly. Nevertheless, Saggiomo (1957) reports that at 273.2 K ($P_{\text{vap}} = 0.047$ bar) C_4N_2 is a solid, whereas at 296 K ($P_{\text{vap}} = 0.187$ bar) it is a liquid (Fig. 20). Thus, the triple point should be located between these two points. We have collected 16 experimental measurements ranging from 147 K (3.6×10^{-10} bar) to 273.2 K (4.72×10^{-2} bar; Saggiomo, 1957; Khanna et al., 1990; Benilan, 1995; Table 3). Khanna et al. (1990) specify that their measurements are reliable only within a factor of 2.5–3. Lide (2006) reports no data concerning the vapour pressure of C_4N_2 . We have calculated an empirical relation for the whole data set that reproduces the experimental measurements with an accuracy of $\pm 90\%$ (Tables 4 and 5 and Fig. 20). Nevertheless, the slopes of the data sets of Benilan (1995) and Khanna et al. (1990) are significantly different and lead to individual extrapolations differing by up to an order of magnitude. Thus, the empirical relation that we propose is uncertain. New measurements are required to more precisely define the vapour pressure of C_4N_2 .

3.2.30. N_2 (nitrogen)

The triple point of N_2 is located at 63.14 ± 0.06 K and 0.1255 ± 0.0005 bar (Table 2). We have collected 38 experimental measurements ranging from 21.2 K (1.467×10^{-13} bar) to 60.67 K (7.367×10^{-2} bar; Giauque and Clayton, 1933; Borovik et al., 1960; Frels et al., 1974; Table 3 and Fig. 21). Lide (2006) reports data for the sublimation and evaporation equilibria in agreement with previous experimental measurements. We have excluded the data of Atake and Chihara (1974), which are too different from the others. Solid N_2 has two different crystalline forms, α and β , with the temperature of solid–solid transition at 35.61 ± 0.05 K (Giauque and Clayton, 1933). Brown and Ziegler (1979) have proposed thermodynamic relations for each phase, which surprisingly do not accurately reproduce the experimental measurements. The relation for the α phase ($T < 35.6$ K) overestimates the experimental measurements by $\sim 30\%$, whereas that for the β phase ($T > 35.61$ K) underestimates the experimental measurements by $\sim 20\%$. We have thus computed our own thermodynamic relations for each phase (Table 4 and 5) using values of heat capacity of N_2 ices measured by Giauque and Clayton (1933) and the constant value $(7/2)R$ for the heat capacity

of the gas, which corresponds to the value of a perfect diatomic gas. Our relations for the α and β phases reproduce the experimental measurements with more accuracy ($\pm 28\%$ and $\pm 2.7\%$, respectively) than those of Brown and Ziegler (1979). In the astrophysical literature, empirical relations have been proposed by Yamamoto et al. (1983) and Huebner et al. (2006). Yamamoto et al. (1983) overestimate all measurements below 35 K and Huebner et al. (2006) reproduce experimental measurements with an accuracy of $\pm 40\%$. Thus, we recommend the use of both our thermodynamic relations.

3.2.31. NH_3 (ammonia)

The triple point of NH_3 is located at 195.41 ± 0.01 K and 0.0609 ± 0.0003 bar (Table 2). We have collected only 15 experimental measurements ranging from 162.39 K (1.27×10^{-3} bar) to 195.36 K (0.0608 bar; Karwat, 1924; Overstreet and Giauque, 1937; Table 3 and Fig. 22). The different dataset reported by Lide (2006) is not in agreement with them for T of about 170 K. The heat capacity of the solid has been measured by Overstreet and Giauque (1937) between 15 and 195 K and the heat capacity of the gas can be calculated down to 15 K thanks to the internal partition function given in the HITRAN database (Rothman et al., 2005), which has been extrapolated using a $T^{1.5}$ dependence for $T < 70$ K. From these data we have computed a thermodynamic relation from 15 K to the triple point temperature (Tables 4 and 5 and Fig. 22). This relation reproduces experimental data with an

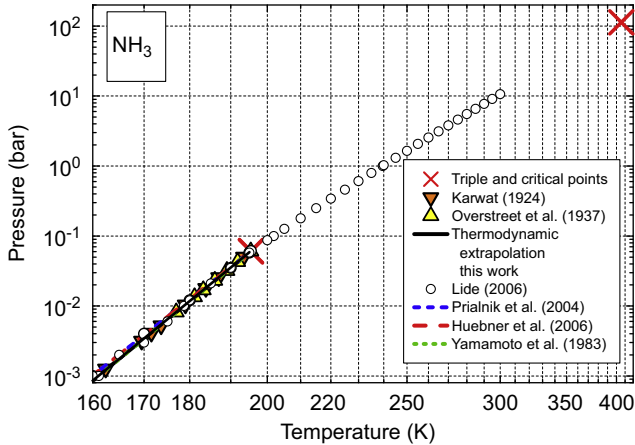


Fig. 22. Sublimation and evaporation pressures of NH_3 as a function of temperature (reciprocal scale $1/T$).

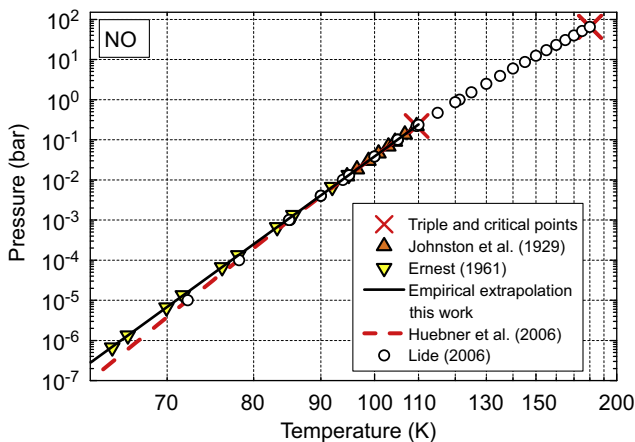


Fig. 23. Sublimation and evaporation pressures of NO as a function of temperature (reciprocal scale $1/T$).

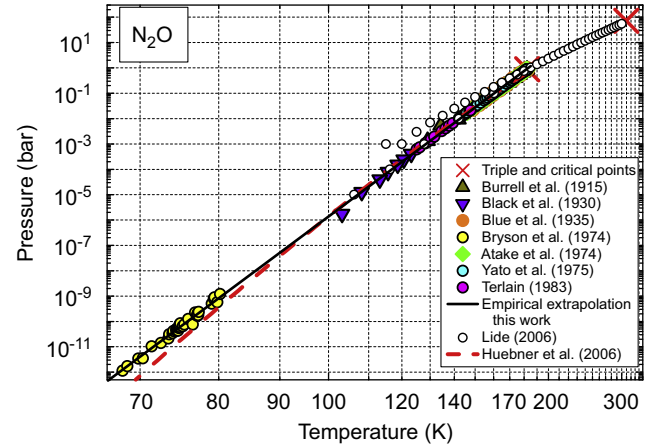


Fig. 24. Sublimation and evaporation pressures of N_2O as a function of temperature (reciprocal scale $1/T$).

accuracy of $\pm 4.2\%$. In the astrophysical literature, empirical relations have been proposed by Yamamoto et al. (1983), Prialnik et al. (2004), and Huebner et al. (2006). The relation proposed by Yamamoto et al. (1983) is quasi-identical to our empirical relation. The relations proposed by Prialnik et al. (2004) and Huebner et al. (2006) overestimate all the measurements and their slopes are too small to be compatible with the values of heat capacities. Thus, we recommend the use of our thermodynamic relation.

3.2.32. NO (nitric oxide)

The triple point of NO is located at 109.5 ± 0.1 K and 0.219 ± 0.001 bar (Table 2). We have collected 18 experimental measurements ranging from 64.85 K (6.66×10^{-7} bar) to 109.49 K (0.219 bar; Johnston and Giauque, 1929; Ernest, 1961; Table 3). The data reported by Lide (2006) are not in accordance with the experimental measurements for $T < 80$ K (Fig. 23). The heat capacity of the NO ice has been measured between 15.6 and 108.4 K by Johnston and Giauque (1929) and the heat capacity of the gas can be calculated for $T > 70$ K thanks to the internal partition function given in the HITRAN database (Rothman et al., 2005). However, as some experimental measurements of the sublimation pressure are available for T lower than 70 K, the thermodynamic relation that we can compute did not provide any useful extrapolation of the data. So we have performed an empirical polynomial interpolation of the experimental measurements in the range $64.9 \text{ K} < T < 109.5 \text{ K}$, and a simple empirical extrapolation valid for $T < 64.9 \text{ K}$ (Tables 4 and 5). The first relation reproduces the experimental data with an accuracy of $\pm 1\%$ over the whole range but only $\pm 3\%$ for the second. In the astrophysical literature, an empirical relation has been proposed by Huebner et al. (2006), which underestimates by a factor of about 2 the experimental measurements at $T < 75 \text{ K}$ (Fig. 23). Thus, we recommend the use of our empirical interpolation relation for $64.9 \text{ K} < T < 109.5 \text{ K}$ and our empirical extrapolation for $T < 64.9 \text{ K}$.

3.2.33. NO_2 (nitrogen dioxide)

The location of the triple point of NO_2 is unknown. No measurement concerning the vapour pressure of NO_2 has been found. Some vapour pressure data attributed to NO_2 are given in the NIST database. Nevertheless, they correspond in fact to vapour pressure of N_2O_4 (dinitrogen tetroxide) measured by Giauque and Kemp (1938). Thus, no NO_2 vapour pressure relation is proposed.

3.2.34. N_2O (nitrous oxide)

The triple point of N_2O is located at 182.30 ± 0.05 K and 0.879 ± 0.001 bar (Table 2). We have collected 180 experimental measurements ranging from 68.1 K (1.14×10^{-12} bar) to 182.29 K (0.88 bar; Burrell and Robertson, 1915b; Black et al., 1930; Blue and Giauque, 1935; Atake and Chihara, 1974; Bryson et al., 1974; Yato et al., 1975; Terlain, 1983; Table 3). The different dataset reported by Lide (2006) is not in agreement for $T < 175$ K (Fig. 24). The heat capacity of solid N_2O has been measured between 15.2 and 180.7 K by Blue and Giauque (1935) and Atake and Chihara (1974) and the heat capacity of the gas can be calculated for $T > 70$ K thanks to the internal partition function given in the HITRAN database (Rothman et al., 2005). However, as the vapour pressure has been measured down to temperatures lower than 70 K, we prefer to interpolate all the collected data (Tables 4 and 5) rather than to compute a thermodynamic relation. This polynomial interpolation can be used for $T > 68.1$ K. For extrapolation at lower temperatures, we have determined a simple empirical relation. Our two relations reproduce the measurements of the vapour pressure with an accuracy of $\pm 50\%$, which reflects the dispersion of the measurements at low temperature. The relation proposed by Huebner et al. (2006) strongly underestimates the vapour pressure measurements for $T < 90$ K (Fig. 24); thus we recommend our empirical interpolation relation for $T > 68$ K and our empirical extrapolation relation for $T < 68$ K.

3.2.35. $HNCO$ (isocyanic acid)

The location of the triple point of $HNCO$ is unknown. No measurements concerning the vapour pressure of $HNCO$ have been found. Thus, no vapour pressure relation is proposed.

3.2.36. NH_2CHO (formamide)

The triple point of NH_2CHO is located at 275.66 ± 0.8 K and its pressure is unknown (Table 2). Lide (2006) reports data only for the evaporation equilibrium. No data concerning the sublimation equilibrium have been found.

3.2.37. H_2S (hydrogen sulphide)

The triple point of H_2S is located at 187.57 ± 0.15 K and 0.229 ± 0.008 bar (Table 2). We have collected only 5 experimental measurements ranging from 164.9 K (3.21×10^{-2} bar) to 182.8 K (0.161 bar; Giauque and Blue, 1936) (Table 3). The different dataset reported by Lide (2006) is not in agreement at low temperatures (Fig. 25). H_2S exists in three different crystalline

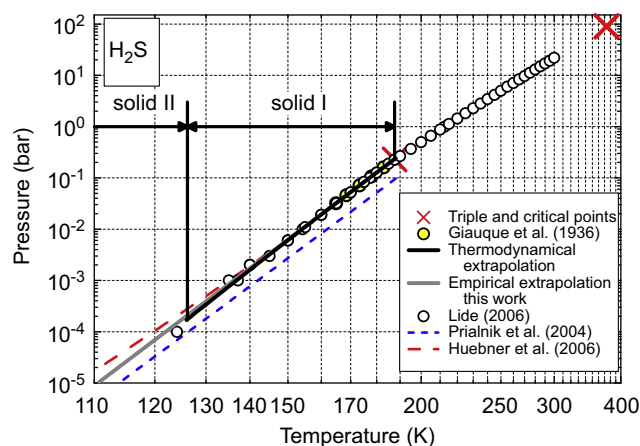


Fig. 25. Sublimation and evaporation pressures of H_2S as a function of temperature (reciprocal scale $1/T$). Vertical lines show the temperature of the phase transition (II–I) and of the triple point.

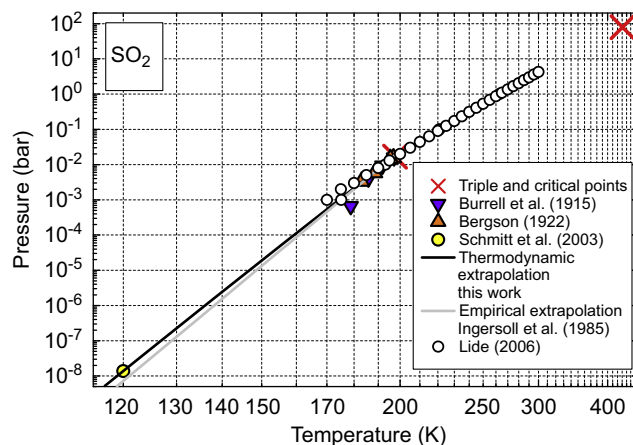


Fig. 26. Sublimation and evaporation pressures of SO_2 as a function of temperature (reciprocal scale $1/T$).

forms. Anderson et al. (1977) showed that phase III transforms to an intermediate phase II at 103.6 K and that a further transition occurs at 126.2 K to phase I. The heat capacity of the solid has been measured between 17 and 186.4 K by Giauque and Blue (1936) and the heat capacity of the gas can be calculated for $T > 70$ K thanks to the internal partition function given in the HITRAN database (Rothman et al., 2005). As a solid–solid transition exists at 126.2 K, we have computed a thermodynamic relation for the range $126.2 \text{ K} < T < 187.57 \text{ K}$ corresponding to phase I and a simple empirical relation for lower T (Tables 4 and 5). These two relations reproduce the experimental data with an accuracy better than $\pm 0.3\%$. New experimental measurements are required to check the validity of these relations at $T < 164.9$ K. In astrophysical literature, empirical relations have been proposed by Huebner et al. (2006) and Pralnik et al. (2004). The relation of Pralnik et al. (2004) underestimates the experimental measurements by a factor of 2.5 (Fig. 25) and the relation of Huebner et al. (2006) reproduces the experimental data with an accuracy of $\pm 3\%$. Thus, we recommend the use of our thermodynamic relation for $127 \text{ K} < T < 187.57 \text{ K}$ and our empirical relation for $T < 127 \text{ K}$.

3.2.38. SO_2 (sulphur dioxide)

The triple point of SO_2 is located at 197.64 ± 0.05 K and 0.0167 ± 0.0001 bar (Table 2). We have collected only 9 experimental measurements ranging from 120 K (1.38×10^{-8} bar) to 197.15 K (1.6×10^{-2} bar; Burrell and Robertson, 1915b; Bergstrom, 1922; Schmitt and Rodriguez, 2003; Table 3). The different dataset reported by Lide (2006) is not in accordance with them for low temperatures and moreover these data for $T < 197.64$ K do not agree with the experimental data (Fig. 26). As can be seen in Fig. 26, the measurement at 179 K of Burrell and Robertson (1915b) seems to be false, and is therefore discarded in our extrapolation relation. The heat capacity of the solid that has been measured between 15.20 and 197.64 K by Giauque and Stephenson (1938) and the heat capacity of the gas have been calculated down to 15 K thanks to the internal partition function given in the HITRAN database (Rothman et al., 2005), which has been extrapolated using a $T^{1.5}$ dependence for $T < 70$ K. From these data, we have computed a thermodynamic relation (Tables 4 and 5), which reproduces the experimental data with an accuracy of $\pm 9.6\%$, reflecting the dispersion of the measurements. In the astrophysical literature, only Ingersoll et al. (1985) have proposed a sublimation relation (Fig. 26), which reproduces the experimental data with an accuracy of $\pm 90\%$ and for which the

slope is too weak to be compatible with the heat capacity values. New measurements of the vapour pressure of solid SO_2 are required to improve the proposed relations. We recommend the use of our thermodynamic relation.

3.2.39. CS_2 (carbonyl disulphide)

The triple point temperature of CS_2 is 161.45 ± 0.40 K and its pressure has never been directly determined (Table 2). From an extrapolation of the evaporation data of Stull (1947b) and Honig and Hook (1960), we can estimate that the triple point pressure is about $(2 \pm 1) \times 10^{-5}$ bar. Lide (2006) reports data only for evaporation equilibrium and no data concerning sublimation have been found. Thus, no sublimation relation can be proposed.

3.2.40. OCS (carbonyl sulphide)

The triple point temperature of OCS is 134.32 ± 0.20 K and its pressure has never been directly determined (Table 2). From an extrapolation of the evaporation data of Stull (1947b) and Honig and Hook (1960), we can estimate that the triple point pressure is about $(5 \pm 2) \times 10^{-4}$ bar. Lide (2006) reports data only for the evaporation equilibrium and no data concerning the sublimation have been found. Thus, no sublimation relation can be proposed.

3.2.41. H_2SO_4 (sulphuric acid)

The location of the triple point of H_2SO_4 is unknown. Lide (2006) reports data only for the evaporation–liquefaction equilibrium and no measurements concerning the sublimation pressure have been found. Thus, no vapour pressure relation is proposed.

3.2.42. NH_4SH (ammonium hydrogensulphide)

The location of the triple point of NH_4SH is unknown. Stull (1947b) reports data only for the evaporation–liquefaction equilibrium and no measurements concerning the sublimation pressure have been found. Thus, no vapour pressure relation is proposed.

3.2.43. H_2CS (thioformaldehyde)

The location of the triple point of H_2CS is unknown. No measurements concerning the vapour pressure of H_2CS have been found. Thus, no vapour pressure relation is proposed.

3.2.44. Cl_2 (chlorine)

The triple point of Cl_2 is located at 172.12 ± 0.15 K and $(1.39 \pm 0.01) \times 10^{-2}$ bar (Table 2). Honig and Hook (1960) and Lide (2006) report extrapolation for the sublimation equilibrium,

whereas no experimental data concerning the sublimation have been found. Thus, we propose no sublimation relation.

3.2.45. ClS_2 (sulphur chloride)

The location of the triple point of ClS_2 is unknown. No measurements concerning the vapour pressure of ClS_2 have been found. Thus, no vapour pressure relation is proposed.

3.2.46. ClO (monochlorine monoxide)

The location of the triple point of ClO is unknown. No measurements concerning the vapour pressure of ClO have been found. Thus, no vapour pressure relation is proposed.

3.2.47. PH_3 (phosphine)

The triple point of PH_3 is located at 139.41 ± 0.20 K and $(3.6 \pm 0.1) \times 10^{-2}$ bar (Table 2). Lide (2006) reports data for the sublimation equilibrium. Nevertheless, it seems that these data have been obtained by extrapolation of the liquid–gas equilibrium for temperatures below the triple point temperature. No experimental data concerning the sublimation have been found. Thus, we propose no sublimation relation.

3.2.48. AsH_3 (arsine)

The triple point of AsH_3 is located at 156.23 ± 0.02 K and $(2.98 \pm 0.02) \times 10^{-2}$ bar (Table 2). We have collected only 13 experimental measurements ranging from 136.15 K (2.67×10^{-3} bar) to 154.15 K (2.47×10^{-2} bar; Johnson and Pechukas, 1937; Table 3 and Fig. 27). Lide (2006) reports data for the sublimation and evaporation equilibria. Sherman and Giauque (1955) showed the existence of a solid–solid transition occurring at $T=105.5$ K from measurements of the heat capacity of the solid from 15.4 to 151.0 K. Unfortunately, as the internal partition function of gaseous AsH_3 is not reported in the HITRAN database, we cannot calculate the heat capacity of the gas and so no thermodynamic relation is proposed. Thus, we have calculated an empirical extrapolation relation (Tables 4 and 5), which reproduces the experimental data with an accuracy of $\pm 6\%$. No other extrapolation relation has been proposed in the literature, so we recommend the use of this empirical relation.

3.2.49. SiH_4 (silane)

The location of the triple point of SiH_4 is unknown. Nevertheless, Stull (1947b) reports that at $T=94$ K ($P_{\text{vap}}=1.4 \times 10^{-3}$ bar), SiH_4 is a liquid. Thus, the triple point of SiH_4 is located below this point. Lide (2006) reports data only for the evaporation

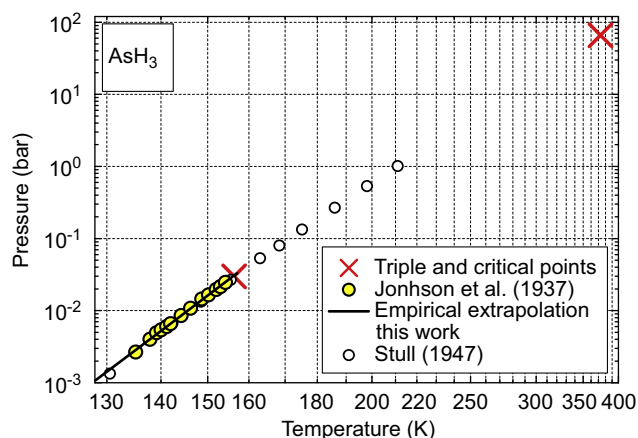


Fig. 27. Sublimation and evaporation pressures of AsH_3 as a function of temperature (reciprocal scale $1/T$).

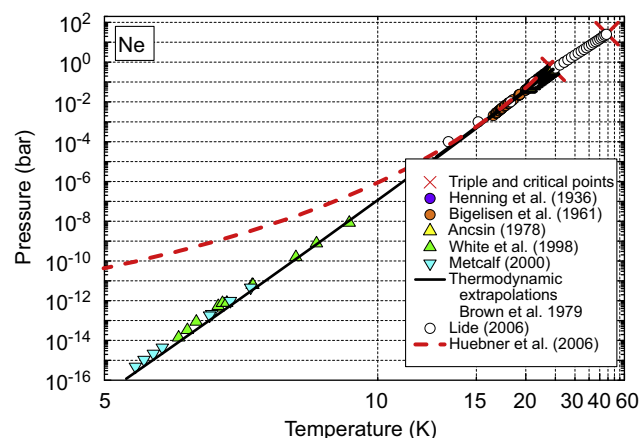


Fig. 28. Sublimation and evaporation pressures of Ne as a function of temperature (reciprocal scale $1/T$).

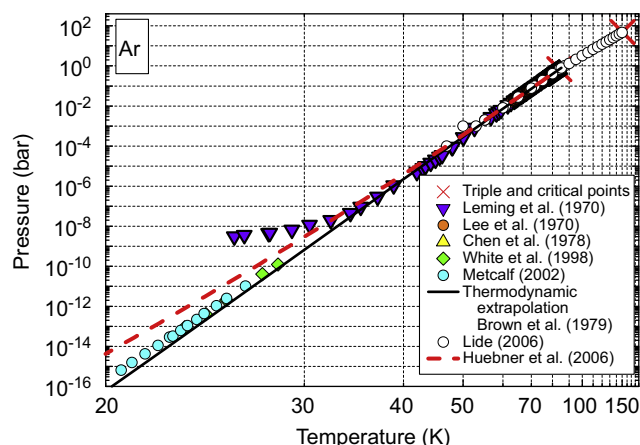


Fig. 29. Sublimation and evaporation pressures of Ar as a function of temperature (reciprocal scale $1/T$).

equilibrium and no data concerning the sublimation have been found. Thus no sublimation relation can be proposed.

3.2.50. Ne (neon)

The triple point of Ne is located at 24.56 ± 0.05 K and 0.433 ± 0.003 bar (Table 2). We have collected 156 experimental measurements ranging from 5.50 K (4.8×10^{-16} bar) to 24.50 K (0.418 bar; Henning and Otto, 1936; Bigeleisen and Roth, 1961; Ancsin, 1978; White et al., 1998; Metcalf, 2002; Table 3). The different dataset reported by Lide (2006) is not in accordance with them for T of about 14 K (Fig. 28). Ne ice exists in only one crystalline form. Nevertheless, Brown and Ziegler (1979) have calculated two thermodynamic relations (Table 4 and 5), first for $3.7 \text{ K} < T < 12.5 \text{ K}$ and then for $12.5 \text{ K} < T < 24.56 \text{ K}$. The second relation reproduces the experimental data with an accuracy of $\pm 9\%$, reflecting the dispersion of the measurements. The very low temperature relation underestimates the experimental measurements of White et al. (1998) and Metcalf (2002) by a factor of 2 at $T < 8 \text{ K}$ (Fig. 28). But we observe a similar problem with the experimental data of White et al. (1998) and Metcalf (2002) concerning argon. Therefore, we consider that these data at extremely low pressure are not accurate. An empirical relation has also been proposed by Huebner et al. (2006), but it overestimates all the measurements, especially below 15 K, where the relation strongly diverges. Thus we recommend the use of the thermodynamic relations proposed by Brown and Ziegler (1979).

3.2.51. Ar (argon)

The triple point of Ar is located at 83.8058 K and 0.68891 ± 0.00002 bar (Table 2). The triple point temperature of Ar was adopted as a defining fixed point on the present temperature scale ITS-90. We have collected 268 experimental measurements ranging from 20.53 K (6.56×10^{-16} bar) to 83.76 K (0.685 bar; Leming and Pollack, 1970; Lee et al., 1970; Chen et al., 1978; White et al., 1998; Metcalf, 2002; Table 3). The dataset reported by Lide (2006) diverges slightly from the experimental measurements for $T < 60 \text{ K}$ (Fig. 29). As for Kr and Xe, the data of Leming and Pollack (1970) present a strong discrepancy at low temperatures; thus we have excluded them below 33 K. Brown and Ziegler (1979) have proposed a thermodynamic relation (Tables 4 and 5), which reproduces experimental data above 55 K with an accuracy of $\pm 5\%$, reflecting the dispersion of the measurements. At lower temperatures, the relation of Brown and Ziegler (1979) underestimates the measurements of

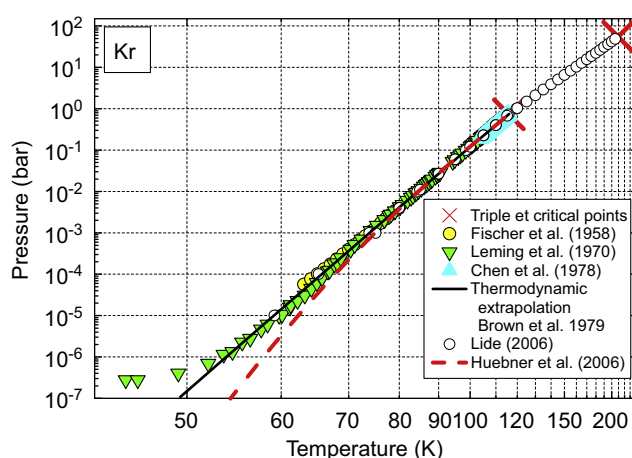


Fig. 30. Sublimation and evaporation pressures of Kr as a function of temperature (reciprocal scale $1/T$).

White et al. (1998) and Metcalf (2002) by a factor of 2 (Fig. 29). We are confident of the relation of Brown and Ziegler (1979) as the thermodynamic relation computed by Tegeler et al. (1999) is almost identical. Thus, we consider, as for Ne, that these extremely low-pressure data may have some systematic error. An empirical relation has also been proposed by Huebner et al. (2006), but it overestimates all the measurements, especially below 50 K. Thus we recommend the use of the thermodynamic relation proposed by Brown and Ziegler (1979).

3.2.52. Kr (krypton)

The triple point of Kr is located at 115.9 ± 0.2 K and 0.7315 ± 0.0006 bar (Table 2). We have collected 161 experimental measurements ranging from 45.13 K (2.8×10^{-7} bar) to 115.63 K (0.72 bar; Fisher and McMillan, 1958; Leming and Pollack, 1970; Chen et al., 1978; Table 3). The dataset reported by Lide (2006) diverges slightly from the experimental measurements for $T < 85 \text{ K}$ (Fig. 30). The measurements of Leming and Pollack (1970) for $T < 55 \text{ K}$ are very doubtful since their slope changes drastically below this temperature (Fig. 30). Moreover, for temperatures below 63 K the data of Leming and Pollack (1970) and Fisher and McMillan (1958) present a discrepancy of about 40%. We decided to exclude the data of Leming and Pollack (1970) below 65 K. The thermodynamic relation computed by Brown and Ziegler (1979) seems to be reliable (Tables 4 and 5 and Fig. 30) as it reproduces the experimental data for $T > 65 \text{ K}$ with an accuracy of $\pm 35\%$, reflecting only the dispersion of the measurements. The empirical relation proposed by Huebner et al. (2006) underestimates all the experimental results below 80 K (Fig. 30). Thus, we recommend the use of the thermodynamic relation computed by Brown and Ziegler (1979).

3.2.53. Xe (xenon)

The triple point of Xe is located at 161.4 ± 0.3 K and 0.816 ± 0.001 bar (Table 2). We have collected 197 experimental measurements ranging from 53.7 K (5.97×10^{-11} bar) to 161.33 K (0.813 bar; Leming and Pollack, 1970; Bryson et al., 1974; Chen et al., 1978; Tessier et al., 1982; Table 3). The dataset reported by Lide (2006) diverges slightly from the experimental measurements for $T < 120 \text{ K}$ (Fig. 31). As for Ar and Kr, the measurements of Leming and Pollack (1970) are very doubtful below 75 K since their slope changes drastically below this temperature (Fig. 31). Thus, we decided to exclude these data below 75 K. The thermodynamic relation computed by Brown and Ziegler (1979) seems to be reliable (Tables 4 and 5) as it reproduces the experimental data for $T > 75 \text{ K}$ with an accuracy of $\pm 20\%$, reflecting the dispersion of the

measurements. The empirical relation proposed by Huebner et al. (2006) strongly underestimates all the experimental results below 120 K (Fig. 31). Thus, we recommend the use of the thermodynamic relation computed by Brown and Ziegler (1979).

3.3. Summary of the bibliographic review

We have searched for experimental measurements of the sublimation pressure of 53 different species.

For 10 of these species (H_2O_2 —hydrogen peroxide, HCOOCH_3 —methyl formate, HNC —hydrogen isocyanide, HNCO —isocyanic acid,

NO_2 —nitrogen dioxide, H_2SO_4 —sulphuric acid, NH_4SH —ammonium hydrogensulphide, H_2CS —thioformaldehyde, ClS_2 —sulphur chloride, and ClO —monochlorine monoxide) the location of the triple point is unknown and no data concerning the sublimation equilibrium have been found. For some of these 10 species, measurements of the vapour pressure over the liquid phase can still be found in Stull (1947a, 1947b) or in Lide (2006).

For 13 other species (H_2CO —formaldehyde, NH_2CHO —formamide, $\text{HOCH}_2\text{CH}_2\text{OH}$ —ethylene glycol, $\text{CH}_2=\text{C}=\text{CH}_2$ —allene, C_3H_6 —propene, C_3H_8 —propane, CH_3CN —acetonitrile, $\text{CH}_3\text{CH}_2\text{CN}$ —propanenitrile, CS_2 —carbon disulphide, OCS —carbonyl sulphide, Cl_2 —chlorine, PH_3 —phosphine, and SiH_4 —silane), the temperature of the triple point is known and in some cases the pressure of the triple point has also been measured. For these 13 molecules, no data concerning the sublimation equilibrium have been found, whereas results concerning the evaporation equilibrium can be found in Stull (1947a, 1947b) and in Lide (2006).

For the 30 remaining species, we have compiled 1842 experimental measurements of their sublimation pressure ranging from 4.8×10^{-16} to 5.2 bar. The intrinsic volatility of these species can be compared in Fig. 32, where the data have been summarized. Taking into account their abundances, their volatility in the solar nebula or in comet nuclei can be compared, as Yamamoto (1985) and Crovisier (1999) did. For species having very different abundances, their order of volatility can be reversed. Nevertheless, Yamamoto (1985) and Crovisier (1999) find qualitatively same results as we do. For 8 of these 30 species, the sublimation pressure has been measured only at relatively high pressures ($> 10^{-3}$ – 10^{-2} bar), whereas it is possible to directly measure absolute pressures as low as 10^{-7} bar with a precision of 1%, thanks to variable-capacitance sensors (Baratron[®], Barocell[®]). Moreover, for 3 simple hydrocarbons the

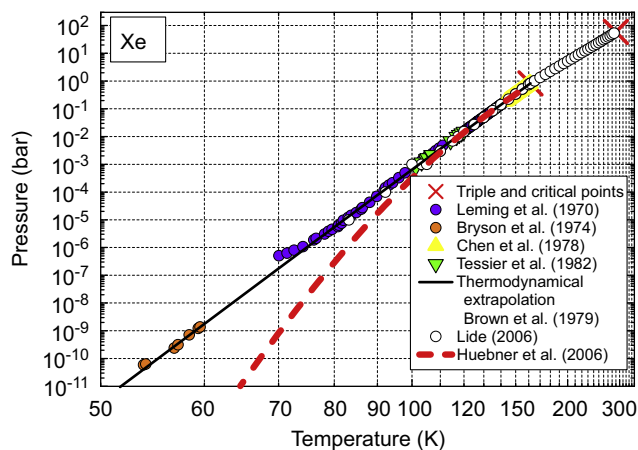


Fig. 31. Sublimation and evaporation pressures of Xe as a function of temperature (reciprocal scale $1/T$).

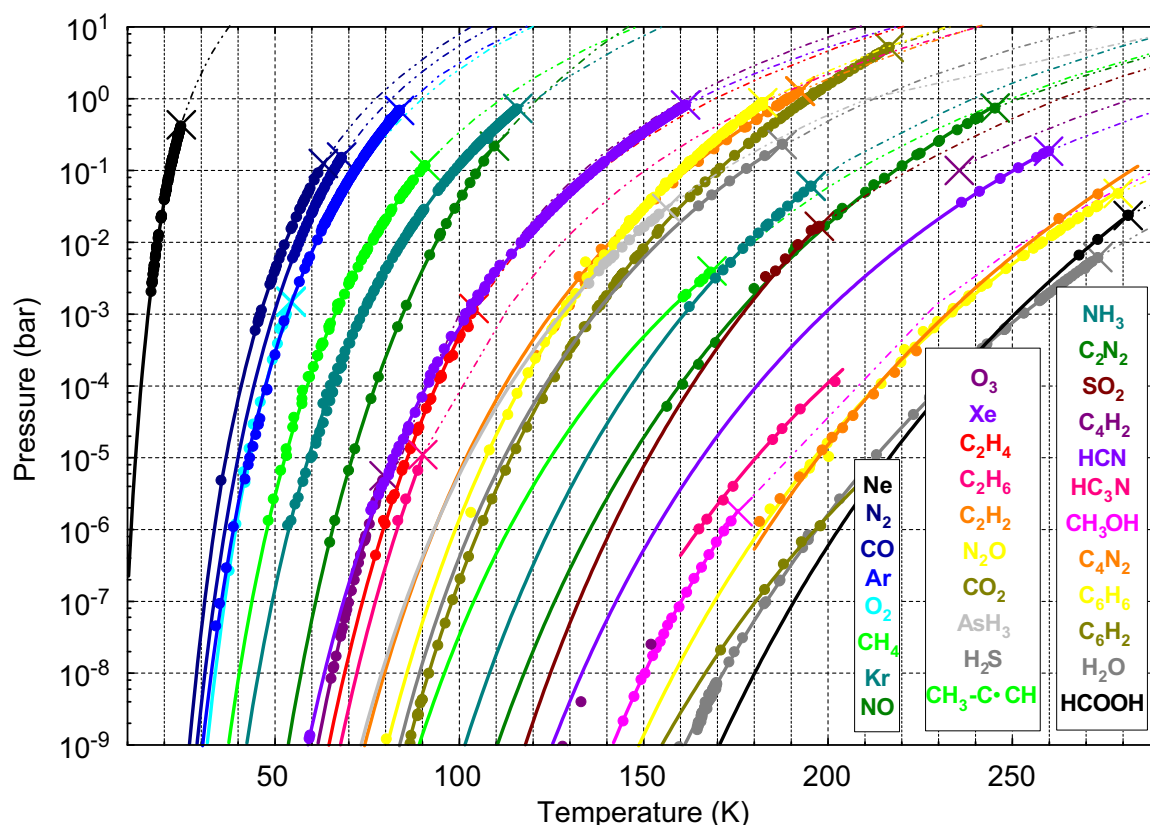


Fig. 32. Sublimation pressure as a function of temperature for the 30 species for which we have found some experimental data. Solid circles and the crosses represent the measurements and the triple points, respectively. The sublimation relations that we recommend (Table 6) are represented as solid lines, whereas dashed lines represent the liquid–gas equilibrium of these species derived from the compilations of Stull (1947a, 1947b) or Lide (2006).

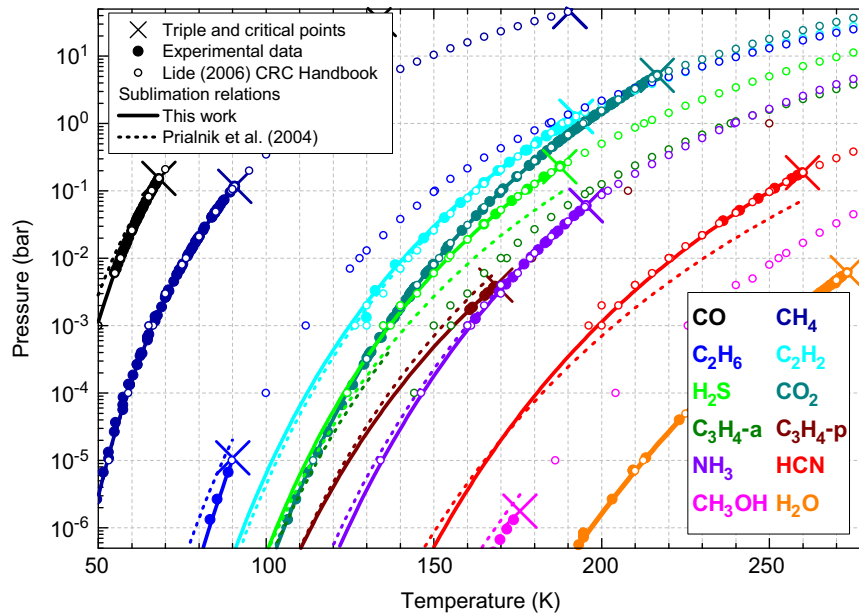


Fig. 33. Comparison between the sublimation relations that we propose and those of Prialnik et al. (2004): CO, CH₄, C₂H₆, C₂H₂, H₂S, CO₂, CH₂=C=CH₂, CH₃-C≡CH, NH₃, HCN, CH₃OH, and H₂O.

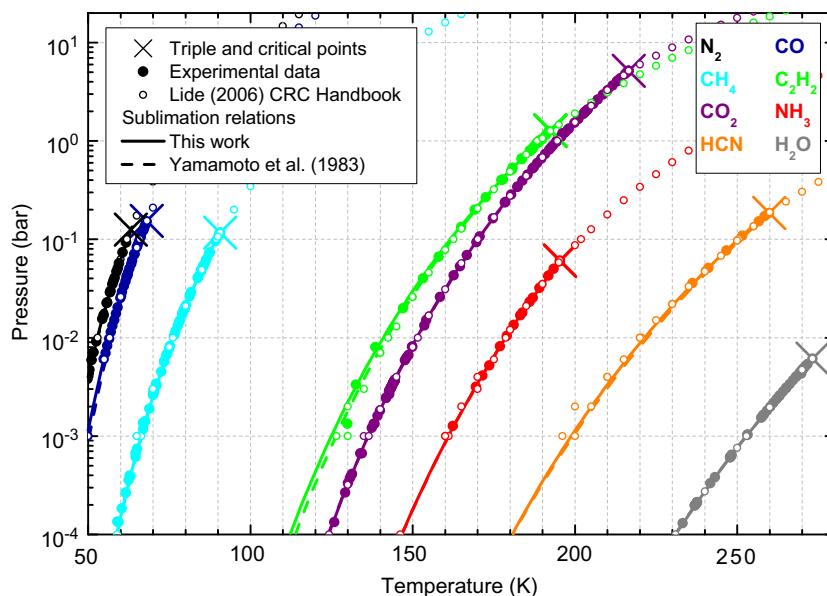


Fig. 34. Comparison between the sublimation relations that we propose and those of Yamamoto et al. (1983): N₂, CO, CH₄, C₂H₂, CO₂, NH₃, HCN, and H₂O.

existing experimental data are either very doubtful (C₄H₂) or not in agreement (C₂H₂ and C₂H₄). Thus, we consider that for 18 of these 30 species, the experimental measurements of their sublimation pressure should be improved. These 18 species are: CO—carbon monoxide, HCOOH—formic acid, CH₄—methane, C₂H₂—acetylene, C₂H₄—ethylene, C₂H₆—ethane, CH₃-C≡CH—propyne, C₄H₂—butadiyne, C₆H₂—hexatriyne, C₆H₆—benzene, HCN—hydrogen cyanide, HC₃N—propionitrile, C₂N₂—cyanogen, NH₃—ammonia, NO—nitric oxide, H₂S—hydrogen sulphide, SO₂—sulphur dioxide, and AsH₃—arsine. The 12 remaining species for which we have found some reliable experimental sublimation curve at pressures lower than 3×10^{-7} bar are: H₂O—water, O₂—oxygen, O₃—ozone, CO₂—carbon dioxide, CH₃OH—methanol, C₄N₂—butynedinitrile, N₂—nitrogen, N₂O—nitric oxide, Ne—neon, Ar—argon, Kr—krypton, and Xe—xenon. For all of these 30 species (except

CH₃-C≡CH, C₄H₂, and C₆H₂), we have proposed either thermodynamic or empirical sublimation relations, allowing us to calculate the sublimation pressure of these molecular solids at any temperatures.

3.4. Comparison between our results and the sublimation relations used in the astrophysical literature

In this section, we compare some empirical sublimation relations proposed by Prialnik et al. (2004), Yamamoto et al. (1983), and Huebner et al. (2006), widely used in the astrophysical literature and in our own work. As already discussed in Section 3.2, all the proposed relations, either thermodynamic or empirical (Tables 4 and 5), are more accurate than those proposed

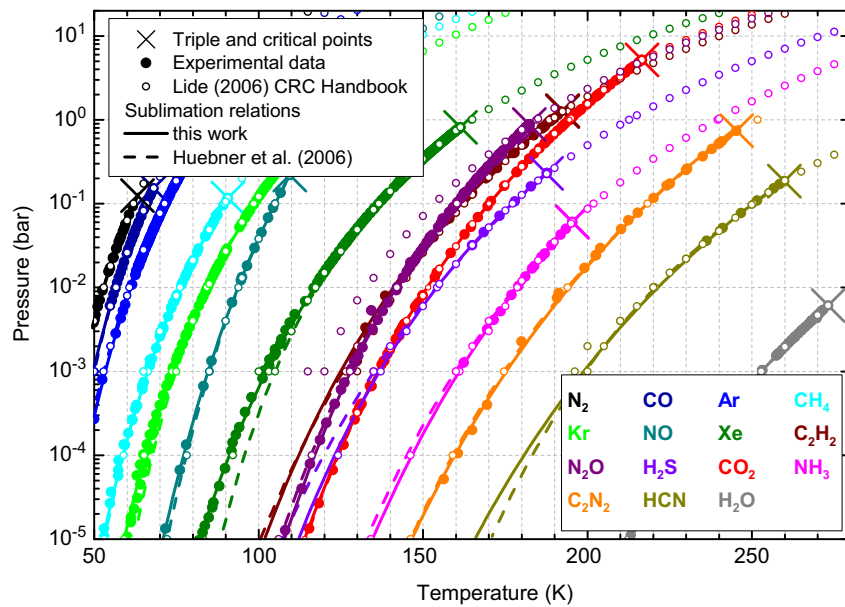


Fig. 35. Comparison between the sublimation relations that we propose and those of Huebner et al. (2006): N_2 , CO, Ar, CH_4 , Kr, NO, Xe, C_2H_2 , N_2O , H_2S , CO_2 , NH_3 , C_2N_2 , HCN, and H_2O .

by the above-mentioned authors. All the former relations are extrapolations of the values published in Lide (2006) or in previous editions of the CRC Handbook of Chemistry and Physics. As we saw in Section 3.2, the data reported by Lide (2006) are not always accurate and can lead to erroneous extrapolations. Yamamoto et al. (1983) and Prialnik et al. (2004) have used relations of the form $\ln(P)=A+B/T$, whereas Huebner et al. (2006) used a relation of the form $\log(P)=A+B/T+C\log(T)+DT$. Figs. 33–35 compare with the experimental data, the relations that we propose and those of Prialnik et al. (2004), Yamamoto et al. (1983), and Huebner et al. (2006), respectively.

As can be seen in Fig. 33, among the 12 sublimation relations proposed by Prialnik et al. (2004), 7 display the same slope as in the evaporation equilibrium curve (CO , C_2H_6 , H_2S , $\text{CH}_2=\text{C}=\text{CH}_2$, $\text{CH}_3-\text{C}\equiv\text{CH}$, HCN, and CH_3OH). Thus, all these relations are false and should not be used at temperatures lower than the triple point temperature. The most problematic case is that of CO. Prialnik et al. (2004) used the relation that was originally proposed by Fanale and Salvail (1990) and which is used in most of the models simulating the activity of comet nuclei. For example, at 54 K this relation overestimates the sublimation pressure by a factor of two, and this error increases with decreasing temperature. Thus, using the relation that we propose could probably significantly change the results concerning the activity of comet nuclei at heliocentric distances greater than 4 AU. The other relations proposed by Prialnik et al. (2004) (for CH_4 , C_2H_2 , CO_2 , NH_3 , and H_2O) reproduce fairly well the experimental sublimation data at temperatures close to the triple point. Nevertheless, the accuracy of these relations decreases at lower temperatures. For example the sublimation pressure of CO_2 is already overestimated by a factor of two around 80 K.

The extrapolation relations proposed by Yamamoto et al. (1983) are quite accurate, but the slopes for CO, C_2H_2 , NH_3 , and HCN are not perfect, leading to moderate discrepancies at low temperatures (Fig. 34).

The relations proposed by Huebner et al. (2006) reproduce the data published in Lide (2006) very well. However, for several species, experimental data exist at temperatures lower than the lowest point published in Lide (2006). Thus, most of the relations proposed by Huebner et al. (2006) do not reproduce accurately the experimental data at low temperature. Moreover, as already

discussed in Section 3.1, the empirical expression used by Huebner et al. (2006) does not allow us to control the slope of the curve at low temperature. Thus, as shown in Fig. 35, some of the extrapolation relations strongly diverge from the experimental data. This is particularly the case for Kr and Xe below 10^{-3} bar.

In conclusion, most of the relations that have been used in the astrophysical literature are quite accurate for temperatures close to the triple point. But, at lower temperatures, the accuracy of these relations decreases significantly and in some cases the error can reach one order of magnitude or more. The relations that we propose in Tables 4 and 5 are more accurate than all the sublimation relations published so far in the astrophysical literature, particularly at low temperature. Indeed, we have compiled a large number of experimental measurements at temperatures lower than the lowest point published in Lide (2006) and we have used, when possible, accurate thermodynamic calculations to extend the sublimation relations at even lower temperatures. As they are based on measurements of the heat capacity, the slopes of the thermodynamic relations that we propose are much more accurate.

4. Conclusion

We performed an extensive bibliographic search of ‘all’ published experimental measurements of the sublimation pressure for 53 different species. For 23 of these species, no experimental measurement has been found. For the 30 remaining species, we have compiled 1836 sublimation pressure measurements ranging from 4.8×10^{-16} to 5.2 bar, and we propose extrapolation relations at low temperatures that can be easily used for applications in different astrophysical environments, such as solar nebula, comet nuclei, or interstellar and circumstellar clouds. For 16 of these 30 species, we have calculated sublimation pressures at low temperature from thermodynamic formulae using measurements of heat capacities of the ice and the gas. Thus, the slope of these extrapolations should be very accurate. For the other species, we are able to propose empirical extrapolations that are more accurate than those generally used in the astrophysical literature.

This work concerns only pure ices. However, in several astrophysical environments, especially at very low temperature, the gases condense as a molecular mixture of different compounds. Good examples are the comet nuclei, which mostly contain water but are also made of numerous mixed compounds. Moreover, if the species are trapped in amorphous water ice, the mixture is not in thermodynamic equilibrium and it is impossible to define an 'equilibrium vapour pressure'. The species included at low temperature in the amorphous water ice structure will be released irreversibly during heating and crystallisation of the amorphous water ice (Bar-Nun et al., 1985; Schmitt et al., 1989). In the case when several compounds are mixed at equilibrium at the molecular level in ices, Henry's law can be used to estimate the vapour pressures of compounds having a very low abundance in the mixture, and Raoult's law for the dominant compound. Raoult's law states that the partial vapour pressure for the dominant compound of a condensed mixture is equal to the product of its molar fraction in the condensed mixture and its vapour pressure in the pure state. In the case of a non-ideal mixture, this law is valid only for molar fractions close to unity. However, for ideal mixtures (e.g. mixtures of compounds having similar chemical structures) it is valid for all possible values of molar fraction. Henry's law states that the vapour pressure of a minor compound of a binary mixture is proportional to its molar fraction in the condensed mixture. However, Henry's coefficient needs to be measured as a function of temperature for each couple of species. For ideal mixtures, a frequent situation with similar species, Raoult's law can also be used. This is the case in particular for isotopic species. If the isotopic composition of an ice is not the 'natural abundance' (all the sublimation curves presented in this paper are for ices with the natural isotopic abundance), then its vapour pressure can be corrected from that of the 'natural ice' by knowing this new isotopic composition. Both laws can be applied for several important planetary cases where one species strongly dominates the ice composition. For example part of the surfaces of Pluto and Triton are mainly composed of N_2 ice (Raoult's law) with minor inclusions (<1%) of CH_4 and CO (Henry's law) in its crystalline structure.

It should be emphasized that there are also a large number of astrophysical situations where the 'mixed ice' is in fact made of a mixture of several types of pure ice grains, each with their own sublimation equilibrium. The bulk of the south permanent cap and the seasonal condensates on Mars, made of pure CO_2 ice mixed with trace amounts of pure H_2O ice grains (and dust) at the grain level, belong to this category of mixture.

Finally various species can also be trapped in clathrate hydrate, a water ice structure containing gas species within cages (Sloan and Koh, 2008). This type of structure, involving volatile species and water, is in thermodynamic equilibrium with the gas under specific conditions. The vapour pressure of a clathrate hydrate, (almost) always lower than that of the pure ice at a given temperature (Lunine and Stevenson, 1985), cannot be determined from the values of the pure ice and thus should be directly measured, or calculated with specific thermodynamic theories. In a number of astrophysical situations, these compounds may compete with pure or mixed ices, and may thus control the gas pressure and composition, and the exchanges between the gas and solid phases. Thus, a compilation of the existing data and an extrapolation of their equilibrium curves should also deserve strong attention.

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